

Reviewed Preprint

v1 • August 5, 2025

Not revised

Reviewed Preprint

v2 • April 2, 2026

Revised by authors

For correspondence:

nobuyoshi@matsumoto.ac

yuji@ikegaya.jp

Competing interests:

No competing interests declared

Funding:

See page 17

Reviewing editor:

Julijana Gjorgjieva, Technical University of Munich, Germany

© 2025, Yamashiro et al. This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Enhanced Tactile Coding in Rat Neocortex Under Darkness

Kotaro Yamashiro¹, Shiyori Tanaka¹, Nobuyoshi Matsumoto^{1,2} , Yuji Ikegaya^{1,2,3} 

¹Graduate School of Pharmaceutical Sciences, The University of Tokyo, Tokyo, Japan • ²Institute for AI and Beyond, The University of Tokyo, Tokyo, Japan • ³Center for Information and Neural Networks, National Institute of Information and Communications Technology, Osaka, Japan

eLife Assessment

This study presents **valuable** findings and employs modern analytical approaches on how transient absence of visual input (darkness) affects tactile encoding in the rat somatosensory cortex (S1). The evidence supporting the authors' claims is **solid**, as population-level neural activity recorded in S1 and decoded by a CNN carries more discriminable texture information in darkness. The underlying basis of this effect remains only partly resolved, however, because it is still unclear which neural features from the CNN drive the decoding and if visual interference is appropriately accounted for, which might confound true neural representational change.

<https://doi.org/10.7554/eLife.106554.2.sa3>

Abstract

Sensory systems are known for their adaptability, responding dynamically to changes in environmental conditions. A key example of this adaptability is the enhancement of tactile perception in the absence of visual input. Despite behavioral studies showing visual deprivation can improve tactile discrimination, the underlying neural mechanisms, particularly how tactile neural representations are reorganized during visual deprivation, remain unclear. In this study, we explore how the absence of visual input alters tactile neural encoding in the rat somatosensory cortex (S1). Rats were trained on a custom-designed treadmill with distinct tactile textures (rough and smooth), and local field potentials (LFPs) were recorded from S1 under light and dark conditions. Machine learning techniques, specifically a convolutional neural network, were used to decode the high-dimensional LFP signals. We found that the neural representations of tactile stimuli became more distinct in the dark, indicating a reorganization of sensory processing in S1 when visual input was removed. Notably, conventional amplitude-based analyses failed to capture these changes, highlighting the power of deep learning in uncovering subtle neural patterns. These findings offer new insights into how the brain rapidly adapts tactile processing in response to the loss of visual input, with implications for multisensory integration.

Introduction

The primary somatosensory cortex (S1) is integral to the encoding of tactile information, processing sensory inputs from various regions of the body (Delhay et al., 2018 ; Di Plinio et al., 2020 ; Piras et al., 2020 ; Serino, 2019 ). This cortical area is crucial for our sense of touch, facilitating the perception and interpretation of sensations such as pressure, vibration, temperature, and pain (Bushnell et al., 1999 ; Luna et al., 2005 ; Moulton et al., 2012 ). These external stimuli are represented by dynamic neural activity in S1, enabling animals to discriminate distinct sensory experiences (Bensmaia et al., 2008 ; Goodwin and Wheat, 2004 ; Koch and Fuster, 1989 ; Salinas et al., 2000 ). Crucially, neural representations in S1 are highly adaptable, shaped not only by feedforward mechanisms but also by attention, motivation, and inputs from other sensory modalities (Butler et al., 2012 ; Driver and Spence, 2000 ; Eimer and

Forster, 2003 [↗](#); Schürmann et al., 2004 [↗](#); Ziegler et al., 2023 [↗](#)). This adaptability facilitates rapid adjustments to changing environmental contexts, supporting survival and optimal sensory processing (Abraira and Ginty, 2013 [↗](#); Dijkerman and de Haan, 2007 [↗](#)).

One prominent example of such adaptability is cross-modal processing, wherein the absence or reduction of visual input enhances tactile abilities. (Bulusu and Lazar, 2024 [↗](#); Hopkins et al., 2017 [↗](#); Nikbakht et al., 2018 [↗](#); Sugiyama et al., 2019 [↗](#)). For instance, prolonged visual deprivation, such as blindness from an early age, results in heightened tactile discrimination abilities (Goldreich and Kanics, 2003 [↗](#); Norman and Bartholomew, 2011 [↗](#); Van Boven et al., 2000 [↗](#); Wong et al., 2011 [↗](#)), driven by cortical reorganization where visual cortical areas are repurposed to support tactile processing (Burton, 2003 [↗](#); Karlen et al., 2006 [↗](#); Sadato et al., 1996 [↗](#)). Notably, however, tactile performance improvements are also observed in sighted individuals during short-term visual deprivation (Facchini and Aglioti, 2003 [↗](#); Kauffman et al., 2002 [↗](#); Pascual-Leone and Hamilton, 2001a [↗](#)). Such individuals demonstrate enhanced tactile discrimination during temporary darkness or blindfolding, indicating that rapid, context-dependent compensatory mechanisms occur even without long-term structural changes (Bola et al., 2017 [↗](#); Boroojerdi et al., 2000 [↗](#); Merabet et al., 2008 [↗](#)). Thus, although tactile enhancement has been studied extensively in the context of long-term sensory deprivation, the neural mechanisms underlying short-term, context-dependent modulation remain unclear.

The neural representation of tactile stimuli within S1 involves complex, distributed activity patterns, posing significant challenges for conventional analyses that primarily measure signal amplitude. Thus, the subtle, high-dimensional neural dynamics that may underlie immediate adaptations to temporary visual deprivation are difficult to detect using traditional methods. Addressing this gap requires refined experimental paradigms and advanced analytical tools to capture how tactile representations reorganize on short timescales.

To examine short-term reorganization in tactile processing, we used a behavioral paradigm using rats trained to walk naturally on a treadmill featuring distinct tactile textures (Yamashiro et al., 2024 [↗](#)). By manipulating visual input (light vs. dark conditions) while recording local field potentials (LFPs) from S1, we aimed to characterize rapid shifts in tactile neural representations. LFPs provide a rich, high-dimensional signal reflecting both synchronous and asynchronous activity across cortical populations. Concurrently, we applied deep learning techniques to decode subtle differences in neural activity patterns that traditional amplitude-based analyses could not detect. We hypothesized that temporary visual deprivation would enhance the neural distinction between tactile stimuli, leading to more differentiated representations of textures in the dark condition.

By investigating these rapid, context-dependent shifts in neural coding, this study provides valuable insights into the brain's capacity for sensory adaptation. Our findings not only shed light on how visual deprivation influences tactile processing in the short term, but also emphasize the potential of deep learning techniques in uncovering nuanced neural reorganization.

Materials and Methods

Animal ethics

Animal experiments were performed with the approval of the Animal Experiment Ethics Committee at the University of Tokyo (approval numbers: P29–7 and P4–15) and according to the University of Tokyo guidelines for the care and use of laboratory animals. These experimental protocols were carried out following the Fundamental Guidelines for the Proper Conduct of Animal Experiments and Related Activities of the Academic Research Institutions (Ministry of Education, Culture, Sports, Science and Technology, Notice No. 71 of 2006), the Standards for Breeding and Housing of and Pain Alleviation for Experimental Animals (Ministry of the Environment, Notice No. 88 of 2006) and the Guidelines on the Method of Animal Disposal (Prime Minister's Office, Notice No. 40 of 1995). While our experimental protocols have a mandate to

humanely euthanize animals if they exhibit any signs of pain, prominent lethargy, and discomfort, such symptoms were not observed in any of the rats tested in this study. All efforts were made to minimize the animals' suffering.

Behavioral paradigm

To record LFPs during natural locomotion, we employed a custom-designed, disk-shaped treadmill with a diameter of 90 cm (Figure 1A [↗](#)). The treadmill's running surface was divided into two halves, each featuring a distinct texture: one side was coated with coarse sandpaper (grain #80) and the other with fine sandpaper (grain #1000). In this setup, the rat was placed on the treadmill and secured with a fabric vest (Figure 1B [↗](#)).

Before the experiment, rats were water-restricted to ensure motivation. A waterspout was positioned in front of each rat, and its output was synchronized with the treadmill's movement (Figure 1C [↗](#)). Specifically, once the treadmill began to move, 30 μ L of water was dispensed from the spout, encouraging the rat to continue walking to receive its water reward.

Each rat was deprived of water until its body weight reached 85% of its baseline weight, then it was trained to walk on the treadmill for four days (one hour per day), with 30 minutes in a light environment (50 lx) followed by 30 minutes in darkness (0 lx). After successful training, the rats were allowed free access to food and water to regain their original body weight before electrode implantation surgery. Following surgical recovery, the rats were water-restricted again and placed on the treadmill to assess cross-modal interactions between the visual and tactile systems. In a single session, LFP recordings were obtained in the light environment trial, followed immediately by the dark environment trial (Figure 1D [↗](#)). For subset of sessions, the order of the light and dark environment was swapped, dark environment trial preceding the dark environment trial. Each trial lasted for approximately 10 minutes. Recording was performed once a day for 4-14 days.

During the trials, the trajectory of the forelimb and onsets when the forelimb came in contact with the floor were automatically identified using a previously developed deep-learning-based method (Figure 1E [↗](#)) (Yamashiro et al., 2024 [↗](#)).

Animal preparation and surgical procedures

LFPs were recorded from eleven 9- to 10-week-old Long-Evans rats (Japan SLC, Shizuoka, Japan) using a custom-designed, 32-channel electrode assembly. This assembly, fabricated from nichrome wires (761500, A-M Systems, WA, USA), targeted the right somatosensory cortex (S1) regions corresponding to the forelimb and hindlimb representations (Figure 1F [↗](#)). Specifically, 18 and 14 electrodes were placed in the forelimb and the hindlimb subregion respectively (Figure 1G [↗](#)). Each electrode tip was platinum-coated to reduce impedance to below 200 k Ω using a nanoZ tester (Plexon, TX, USA).

At the start of the surgical procedure, each rat was anesthetized with 2–3% isoflurane gas. A square craniotomy (2–6 mm posterior and 1–5 mm lateral to bregma) was then created using a dental drill. The electrode assembly was gently lowered through the cranial window to a depth of approximately 1.5 mm beneath the dura, targeting layer IV of S1. Additionally, two stainless steel screws were implanted in the bone above the cerebellum to serve as ground and reference electrodes. The recording device and electrodes were secured to the skull using stainless steel screws and dental cement. Following the surgery, each rat was housed individually in a transparent Plexiglas cage with ad libitum access to food and water for one week to ensure proper recovery.

LFP recordings from S1

LFPs were referenced to ground, digitized at 30 kHz using the OpenEphys recording system (<http://open-ephys.org> [↗](#)) and an RHD 32-channel headstage (C3314, Intan Technologies, CA, USA), then resampled to 10 kHz for subsequent analyses (Yamashiro et al., 2020 [↗](#)). In parallel, video was acquired at 60 Hz using a USB camera module (MCM-303NIR, Gazo, Niigata, Japan), capturing a lateral view of the rat. Each video frame was synchronized with the neural recordings using

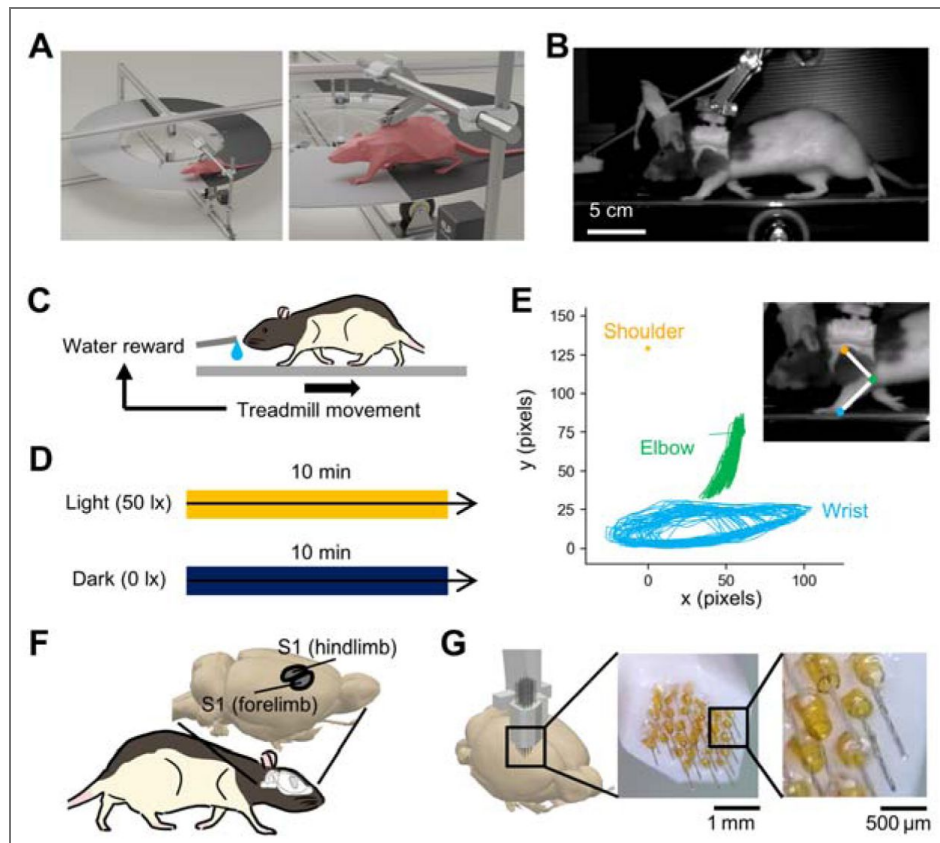


Figure 1. Behavioral paradigm and limb movement assessment with concurrent LFP recordings.

(A) A diagram of the disk-shaped treadmill used in the experiment. One half of the disk is covered with #80 sandpaper, and the other half with #2000 sandpaper. (B) A frame from the video capturing a walking rat from a left-side perspective. (C) The motivation scheme. The rats were water-deprived prior to the experiments. A water port was coupled with the movement of the treadmill so that when the rat walked on the treadmill, the water would come out. This way, the rats were always motivated to walk during the whole session. (D) The experimental protocol, where each rat walked for 10 minutes in light (50 lx) and then for 10 minutes in darkness (0 lx). (E) An example trajectory of the elbow and wrist joints from one session, plotted with the shoulder joint fixed in the coordinate space (shoulder: yellow, elbow: green, wrist: cyan). (F) A schematic illustrating the forelimb and hindlimb subregions of S1. (G) A custom 32-channel electrode array used to record LFPs from these subregions.

strobe signals. For a subset of recordings ($n = 2$ rats), speed sensors were attached to the running disk to monitor locomotor speed, which was simultaneously recorded through the OpenEphys analog input.

Data analysis

Data was analyzed offline using custom-made scripts in Python3. For box plots, the centerline shows the median, the box limits show the upper and lower quartiles, and the whiskers cover the 10–90% percentiles. $PL < 0.05$ was considered statistically significant. All statistical t -tests were two-sided, and the bootstrap method was applied for multiple comparisons.

LFP analysis

To see if there were any differences in LFPs, the LFPs were aligned to the foot strike onsets detected using deep-learning assisted methods. The aligned LFP were then categorized by the trial (light vs. dark) and the floor texture (smooth vs. rough). To analyze the amplitude of the event-related response from the foot strike, a mean trace of the aligned LFP was calculated for each channel from 32 electrodes for each condition. Foot-strike detection and all LFP alignment were intentionally forelimb-locked, because the behavioral paradigm was designed for forepaw texture contact (Figure 1E) and our array oversampled S1-forelimb (18 channels) relative to S1-hindlimb (14 channels; Figure 1G).

A deep neural network for joint decoding of trial conditions and floor texture

A custom deep neural network (DNN) model was implemented to predict both the trial condition (e.g., light vs. dark) and floor texture (e.g., smooth vs. rough) from time-aligned, one-dimensional LFP segments, using the PyTorch framework. Our model architecture was inspired by one-dimensional ResNet-like structures and incorporated multiheaded outputs for simultaneous prediction of two distinct variables (He et al., 2015).

The input to the model consisted of one-dimensional LFP signals, arranged as a tensor with multiple channels (e.g., 32 input channels) over time. The network began by splitting the input into two parallel convolutional pathways. The first pathway (“left” branch) applied sequential convolutional and pooling operations with relatively smaller kernel sizes and strides to incrementally reduce the dimensionality of the signal and extract fine-grained temporal features. Specifically, the model employed a two-stage convolutional process that first passed the input through a one-dimensional (1D) convolution layer with a kernel size of 7, stride of 2, and batch normalization, followed by a max pooling and an additional convolution layer. Both convolutional layers in the left branch used ReLU nonlinearities to facilitate stable and efficient feature extraction.

In contrast, the second pathway (“right” branch) processed the input through a single 1D convolutional layer with a larger kernel size (e.g., 41) and a more aggressive stride (e.g., stride of 8). This pathway captured broader temporal contexts from the input signals. Similar to the left branch, the right branch output was batch-normalized and passed through a ReLU activation function. After these two parallel extractions, the outputs of the left and right branches were concatenated along the channel dimension, forming a combined feature representation that integrated both fine- and coarse-grained temporal information.

The concatenated output was then passed through a max pooling operation, followed by two residual layers that employed 1D convolutional blocks (ResidualBlock) to refine feature representations. These residual layers allowed the network to learn more complex feature hierarchies by facilitating the flow of gradients during training and improving convergence, while also maintaining temporal resolution appropriate for downstream decoding.

Subsequently, the processed features were passed through an average pooling layer to summarize temporal information into a low-dimensional feature vector. This vector was flattened into a one-dimensional representation and then fed into two separate fully connected “heads” (fc_head1 and

fc_head2). Each head was a simple linear layer that provided a scalar output value. By concatenating these outputs, the final layer jointly predicted two target variables from the same underlying features.

In summary, our model combined parallel convolutional branches for initial feature extraction, residual layers for robust representation learning, and multiheaded outputs to facilitate joint prediction of trial conditions and floor textures. The model was implemented in Python using PyTorch, and all parameters were optimized via standard stochastic gradient-based methods. This architecture allowed efficient and robust decoding of environmental conditions from LFP signals in both time and frequency domains.

Training and evaluating the deep neural network

Of the 11 recorded rats, data from the initial two rats were utilized to determine the optimal model architecture. Once the architecture was established, data from the remaining 9 rats were used for training and evaluation. The raw LFP signals were resampled from 30 kHz to 10 kHz and segmented into 800 ms windows centered on the footstrike onset (*e.g.*, 400 ms before and 400 ms after). Each LFP segment was assigned two labels: one for the trial condition (*e.g.*, light vs. dark) and one for the floor texture (*e.g.*, smooth vs. rough).

For each rat, the dataset was shuffled, normalized, and then subjected to 5-fold cross-validation. In this procedure, the data were partitioned into five equal subsets; in each fold, four subsets (80%) were used for training, and the remaining subset (20%) was used for evaluation. Model training was performed for 80 epochs using a batch size of 128 and a learning rate of 10^{-6} . Binary cross-entropy with logits loss (BCEWithLogitsLoss) served as the objective function. These parameters were selected to ensure stable convergence and to reduce overfitting. After the training, the model's performance was evaluated using the evaluation dataset. The confusion matrix was calculated as the number of true positives, false positives, false negatives, and true negatives, aggregated across all predictions in the evaluation set.

Evaluation of cluster separability between light and dark conditions

To examine how texture representations differed between light and dark conditions, we assessed cluster separability of the intermediate features extracted from the deep-learning model. Specifically, after the model was trained, we extracted the output from the penultimate layer (a 912-dimensional feature vector for each trial). Cluster separability was quantified using the silhouette score, which measures the relative distances between within-cluster and between-cluster samples.

Silhouette scores were calculated for the two texture classes (smooth and rough) within each lighting condition, using the Euclidean distance metric applied to the intermediate feature representations. For each trial, the silhouette score s is defined as:

$$s = \frac{b - a}{\max(a, b)}$$

where a is the average distance between a point and all other points in the same cluster (intra-cluster distance), and b is the minimum average distance between a point and all points in the nearest different cluster (inter-cluster distance). The silhouette score, therefore, provides a normalized index of cluster separability, with larger values indicating more distinct clustering. To obtain an animal-level measure, silhouette scores were averaged across cross-validation within each rat.

Explainability analysis using occlusion and integrated gradients

To elucidate the internal decision-making processes of the trained deep neural network (DNN), we employed two established explainability techniques: occlusion and integrated gradients.

Occlusion

Occlusion analysis involves systematically masking specific input features to determine their relative contribution to the model's output (Zeiler and Fergus, 2013 [↗](#)). In the present study, one channel was selectively occluded at a time from the 32-channel LFP input and the sensitivity of each channel was calculated. Sensitivity was defined as the corresponding change in model performance when the specified channel was occluded. By conducting these analyses for each of the nine rats, channel-specific importance scores were obtained and subsequently normalized (z-scored) to facilitate cross-subject comparisons. Channels whose removal yielded a more pronounced decrease in model performance were considered more critical for accurate prediction.

Integrated gradients

Integrated gradients is an attribution method that quantifies the importance of each input feature by integrating the gradient of the model's output with respect to the input, transitioning from a baseline input to the actual input (Sundararajan et al., 2017 [↗](#)). This approach produces class activation maps, enabling the visualization of features most influential for the model's output. Here, integrated gradients were applied to the LFP segments for each rat, and the resulting class activation maps were averaged across subjects and trial conditions. These maps allowed us to identify salient input regions associated with both trial conditions and floor texture.

Results

Stable locomotion across light and dark conditions

To investigate how visual input influences tactile processing in S1, we devised an experimental paradigm where both tactile and visual inputs were independently manipulated. Rats were placed on a disk-shaped treadmill with two distinct sandpaper textures, and LFPs were recorded from walking rats. Each rat walked for 10 min in a light environment (50 lx) and then for 10 minutes in total darkness (0 lx).

To make sure that rat's trajectory was stable across different floor textures and environmental conditions, gait parameters were extracted from the trajectories using deep-learning-based analysis (Figure 2A [↗](#)). From the trajectories, swing duration, stance duration, stride length, and footstrike speed were extracted. All parameters were calculated for each floor textures and environmental conditions. Comparison of all conditions revealed that none of these metrics differed significantly between floor textures or environmental conditions, indicating that overall locomotion remained stable (Figure 2B-E [↗](#), $P > 0.05$, one-way analysis of variance (ANOVA) followed by TukeyL Kramer *post hoc* test, $n = 149, 149, 107$ and 107 trials for smooth-light, rough-light, smooth-dark, and rough-dark, respectively). Locomotor-speed traces recorded from subset of rats. The result indicated that walking velocity was stable across trials, not affected by the lighting conditions (Supplementary Figure 1 [↗](#)). Thus, any differences in neural activity under these conditions are unlikely to be driven by altered motor behavior.

Characteristic of S1 LFP upon forelimb contact

We next analyzed LFPs from S1 using a custom 32-channel electrode array targeting the forelimb and hindlimb subregions (Figure 1F [↗](#), G). LFP traces were first aligned to forelimb contacts with the disk surface. On inspection of a single LFP trace, no apparent event-related response could be observed (Figure 3A [↗](#)). However, aligning each LFP at forelimb contact with the floor, the average trace showed a clear response after the onset (Figure 3B [↗](#)). Across conditions, we observed a clear negative deflection in LFPs that was most pronounced in the forelimb subregion (Figure 3C [↗](#)). The amplitude of the response was significantly larger in all rats, consistent with the topographic specificity of S1 (Figure 3D [↗](#), $P = 1.89 \times 10^{-3}$, $t_{10} = -2.2$, paired *t*-test, $n = 11$ rats). This was expected since the epochs were forelimb-locked, making larger deflections and greater importance of forelimb channels consistent with somatotopy.

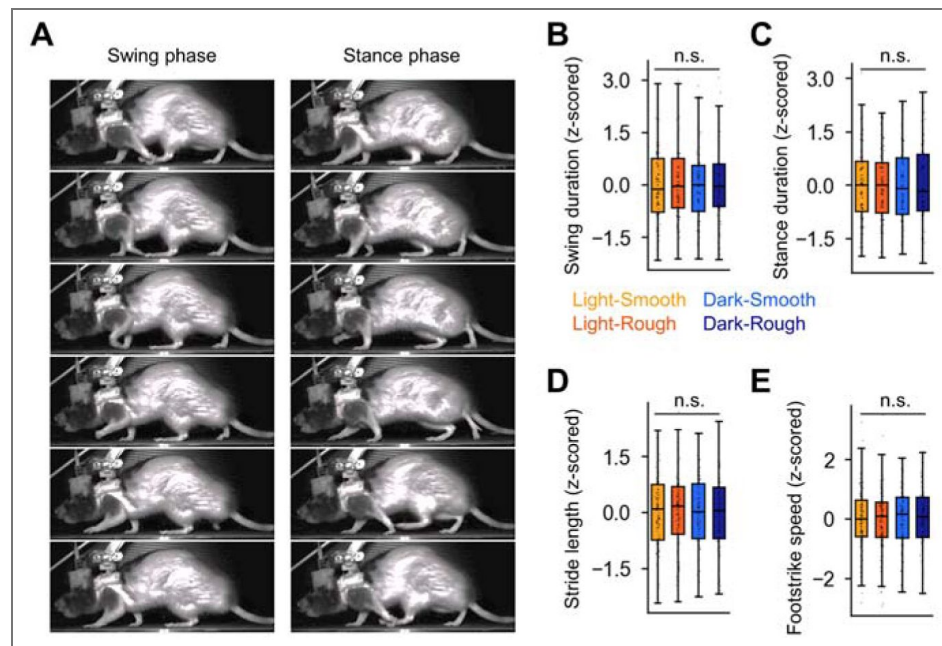


Figure 2. Comparison of gait parameters across textures and environmental conditions.

(A) Swing phase vs. stance phase, illustrated with video frames (*left*: swing, *right*: stance). (B) Normalized swing duration measured for each rat under different textures (smooth vs. rough) and environmental condition (light vs. dark). Light orange and dark orange correspond to the light conditions (smooth, rough), while light blue and dark blue correspond to the dark conditions (smooth, rough). There were no significant differences among trial conditions. (C–E) Stance duration, stride length, and footstrike speed, respectively, under the same conditions as in B. None of these parameters differed significantly across texture types or lighting conditions.

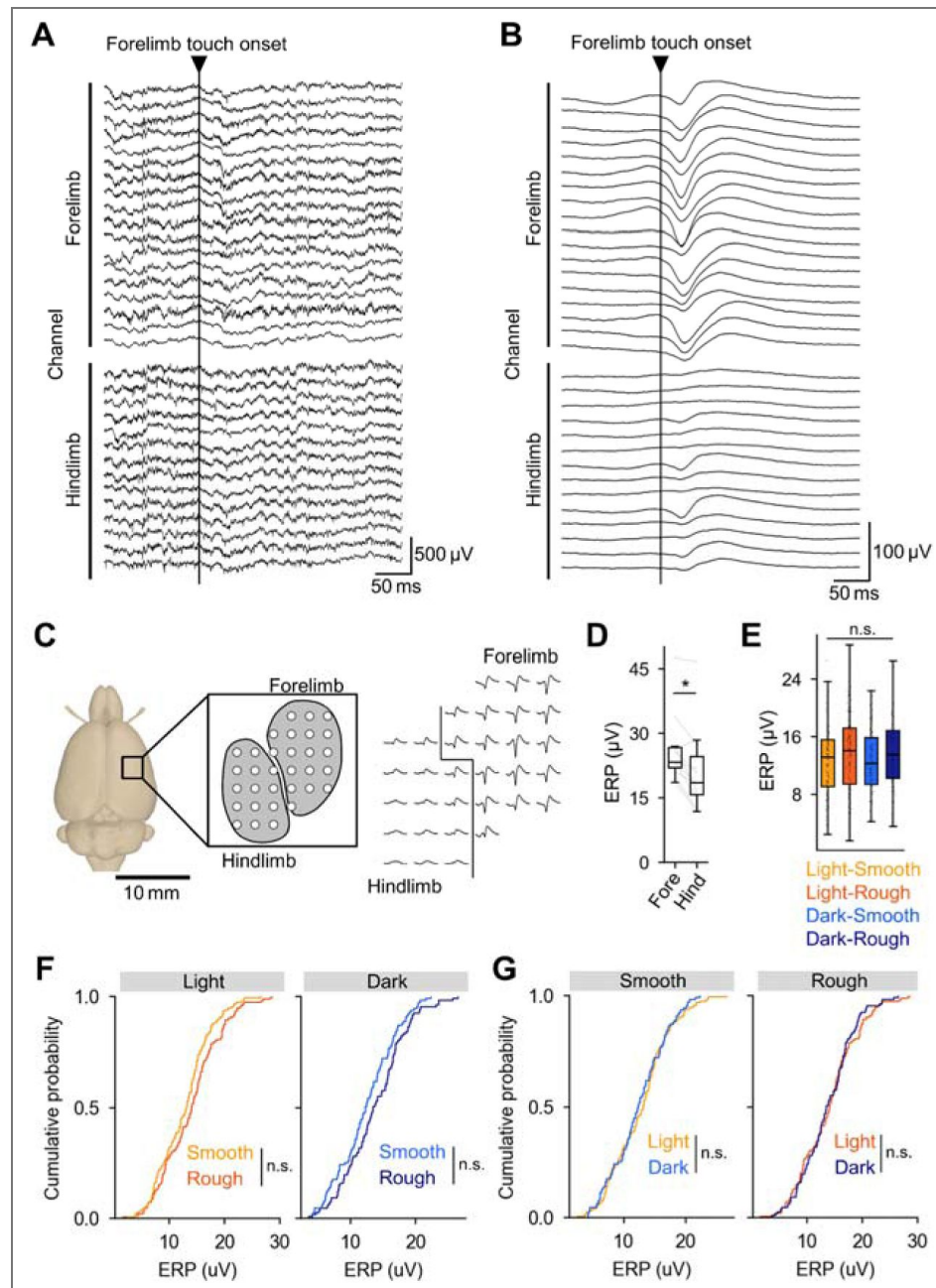


Figure 3. LFP recordings in rat S1 during walking.

(A) A single representative LFP trace aligned to a forelimb contact. (B) An example of an averaged LFP trace from one session, aligned to forelimb contact. (C) The electrode montage and averaged LFP at each electrode. *Left*: The electrode montage showing all 32 recording sites. *Right*: Averaged LFP signals aligned to forelimb contacts with the floor, shown for each electrode depicted in the left panel. (D) Comparison of amplitudes between hindlimb and forelimb subregions, aggregated across all 11 rats. $P = 1.89 \times 10^{-3}$, $t_{10} = -2.2$, paired t-test, $n = 11$ rats. (E) Comparison of averaged amplitudes within the same trial for different floor textures and environmental conditions. $P > 0.05$, one-way analysis of variance (ANOVA) followed by TukeyL Kramer post hoc test, $n = 149, 149, 107$ and 107 trials for smooth-light (light-orange), rough-light (dark-orange), smooth-dark (light-blue), and rough-dark (dark-blue), respectively. (F) Cumulative probability distributions of mean amplitude from each session, compared across different textures. $P = 4.84 \times 10^{-1}$, 8.35×10^{-1} , $D = 1.03 \times 10^{-1}$ and 7.54×10^{-2} for light and dark environments respectively, two-sample KolmogorovL Smirnov test, $n = 149$ and 107 trials from 11 rats for light and dark, respectively. (G) Cumulative probability distributions of mean amplitude from each session, compared across light and dark environments. $P = 1.05 \times 10^{-1}$, 1.83×10^{-1} , $D = 0.14$ and 0.149 for smooth and rough textures respectively, two-sample KolmogorovL Smirnov test, $n = 149$ and 107 trials from 11 rats for light and dark, respectively. Abbreviations: ERP, event-related potential; LFP, local field potential; S1, primary somatosensory cortex.

To see if the event-related responses were affected by the change in floor texture or the environmental condition, the amplitudes were compared. Despite the prominent negative deflection in LFPs at the forelimb contact with the floor, analyses showed no substantial differences in signal amplitude (Figure 3E [↗](#), $P > 0.05$, one-way analysis of variance (ANOVA) followed by TukeyL Kramer *post hoc* test, $n = 149, 149, 107$ and 107 trials for smooth-light, rough-light, smooth-dark, and rough-dark, respectively) when comparing rough vs. smooth textures (Figure 3F [↗](#), $P = 0.48$ and 0.84 , $D = 1.03 \times 10^{-1}$ and 7.54×10^{-2} for light and dark environments respectively, two-sample KolmogorovL Smirnov test, $n = 149$ and 107 trials from 11 rats for light and dark, respectively) and light vs. dark environments (Figure 6G [↗](#), $P = 0.11, 0.18$, $D = 0.14$ and 0.149 for smooth and rough textures respectively, two-sample KolmogorovL Smirnov test, $n = 149$ and 107 trials from 11 rats for light and dark, respectively).

To further investigate, we computed grand-average LFP waveforms across combinations of floor texture (smooth vs. rough) and lighting condition (light vs. dark) (Figure 4A [↗](#)). Visual inspection of the mean waveforms revealed no distinct or prominent features. We then calculated channel-wise correlations between textures within each lighting condition (i.e., light-smooth vs. light-rough; dark-smooth vs. dark-rough). To assess whether lighting modulated texture-related correlations, we subtracted the correlation coefficients obtained in the dark from those in the light and averaged the differences across animals (Figure 4B [↗](#)). The correlation of LFP waveforms between different textures was stronger in the light condition, suggesting that in the dark environment, the average LFP waveform is more distinct. Additionally, we examined channel-channel correlation matrices across all 32 channels for each condition (Figure 4C [↗](#)). While stronger correlations were observed within individual S1 subregions, no systematic differences emerged between lighting conditions or between textures. To probe the frequency domain of the LFP, we computed time-frequency (wavelet) spectrograms across the floor texture and lighting condition combinations. However, no clear differences were observed across these combinations. Taken together, these results suggest that simple amplitude-based metrics may not fully capture subtler or higher-dimensional variations in underlying neuronal activity.

Machine learning-based decoding of tactile and visual information

Given the absence of clear differences in averaged LFP features, we next examined how representations of textures in the LFPs varied across environments using machine learning approaches. We first implemented a support vector machine (SVM) classifier with a radial basis function (RBF) kernel. To reduce dimensionality, principal component analysis (PCA) was applied to the raw LFP traces (Supplementary Figure 2A [↗](#)). The SVM was then trained using 5-fold cross-validation, where, in each fold, 80% of the dataset was used to classify two types of labels: texture (smooth vs. rough) and trial (light vs. dark). Model performance was evaluated on the remaining 20% of the data. The overall classification accuracy was at chance level ($\sim 50\%$), indicating that the PCA-SVM combination failed to extract informative features for texture or lighting condition from the LFP traces (Supplementary Figure 2B [↗](#)).

We next employed a deep learning model, a convolutional neural network (CNN), to classify both floor texture (smooth vs. rough) and trial (light vs. dark). The CNN architecture incorporated parallel convolutional pathways designed to capture both macro- and micro-scale temporal dynamics, followed by residual blocks. The network then branched into dual output layers to jointly predict texture and lighting conditions (Figure 5A [↗](#)).

Similarly to SVM, the model was trained and evaluated using 5-fold cross-validation. In training, the model exhibited stable learning curves (Figure 5B [↗](#)) and also generalized well to held-out test data (Figure 5C [↗](#)). Confusion matrices for both texture and lighting classifications were above chance along the diagonal, demonstrating that the model reliably extracted neural representations of the floor textures from the LFPs (Figure 5D-F [↗](#)).

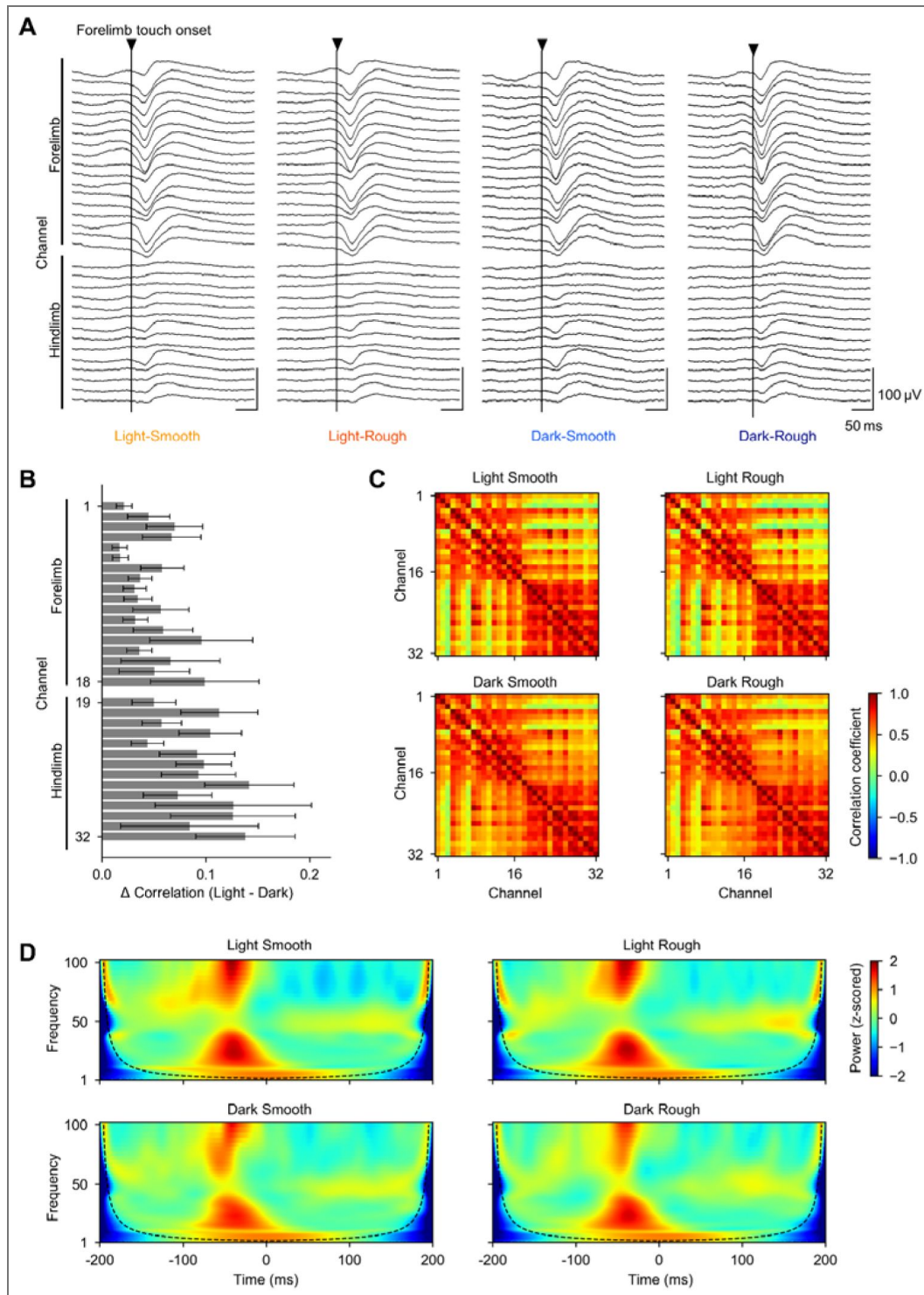


Figure 4. LFP characteristics across textures and lighting

(A) Example averaged LFP traces from one session, aligned to forelimb contact: light-smooth, light-rough, dark-smooth, dark-rough (left to right). (B) Channel correlation difference map (light - dark) computed from average LFP waveforms (Pearson's r). Higher values indicate stronger channel-wise correlations in light relative to dark. (C) Correlation matrices shown separately for each texture and lighting condition. Matrices exhibit similar within-S1-subregion structure, with no clear texture- or light-dependent differences. (D) Example time-frequency (wavelet) spectrogram of LFP power from one session, aligned to forelimb contact. Dotted line indicates the cone of influence. Color scales represent wavelet coefficient magnitude (C) and corresponding power (D).

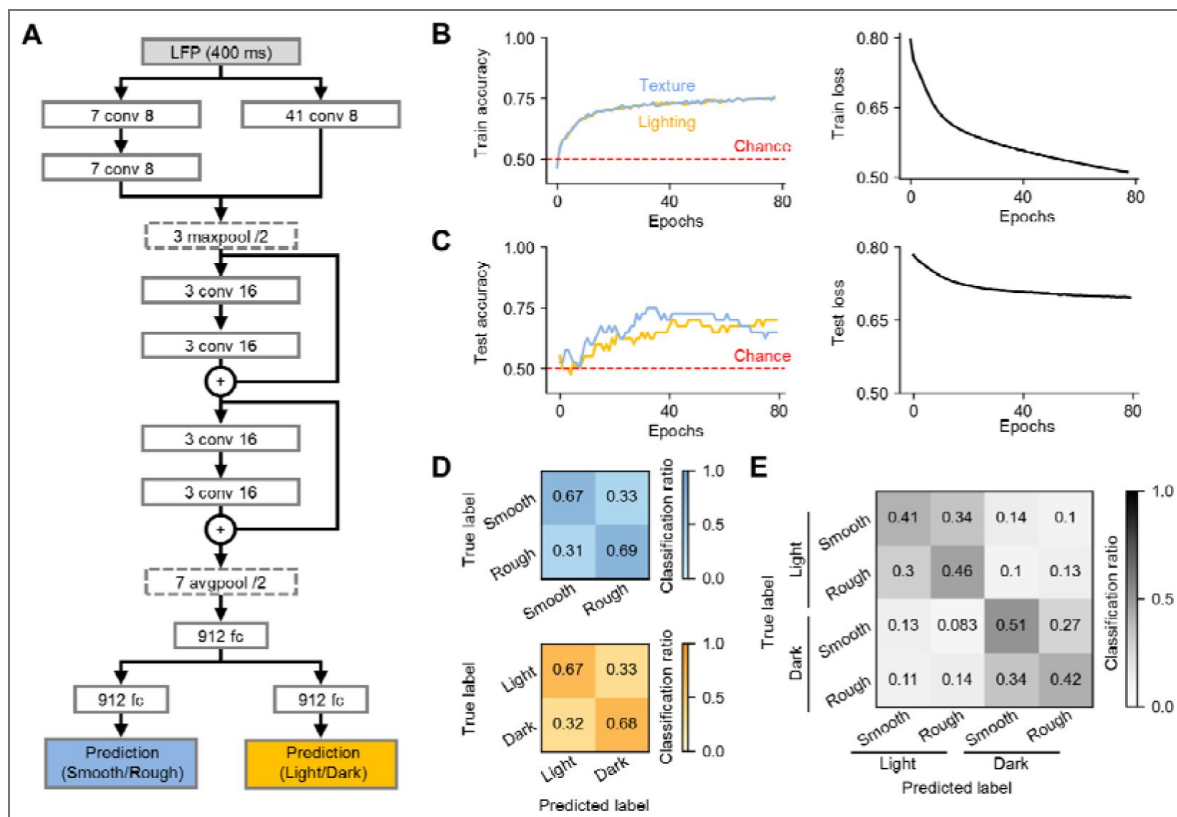


Figure 5. Model-based prediction of texture and environmental conditions from LFP.

(A) The deep learning model architecture. The LFP input is processed through two parallel pathways for macro- and micro-scale feature extraction, followed by residual blocks that feed into two output heads for floor texture (Smooth vs. Rough) and the environmental condition (Light vs. Dark). (B) Training performance for a single representative rat. The left graph shows accuracy curves for texture (blue) and lighting (yellow), and the right graph shows the loss curves. (C) Testing performance for the same rat. The model exhibits good generalization, as accuracy increases and loss decreases on held-out data. (D) Confusion matrix for texture classification for all rats. Values above chance on the diagonal indicate successful texture prediction. Note that all values in the same row add up to 1. (E) Same as D, but for environmental conditions. (F) Combined confusion matrix for texture and trial predictions. The model performs well on both tasks across all rats. Abbreviations: avgpool, average pooling layer; conv, convolutional layer; maxpool, max pooling layer; LFP, local field potential.

Neural representations become more distinct in darkness

To understand how the absence of the visual cue might refine tactile processing, we performed explainability analyses on the model's learned representations (Figure 6C). We extracted a 912-dimensional feature vector from the layer preceding the final outputs, then visualized these high-dimensional embeddings with scatter plots (Figure 6A, B). Clustering analyses using silhouette scores showed that representations of texture were more separated in the dark environment, suggesting that reduced visual input enhances the distinctness of neural dynamics that code tactile stimuli (Figure 6C, $P = 2.31 \times 10^{-2}$, $t_8 = -2.8$, paired t -test, $n = 9$ rats, Supplementary Table 1). To account for trial order, silhouette scores were also calculated in sessions where dark trials preceded light trials. In this case, only an increasing trend was observed, with no significant difference detected (Supplementary Figure 3A, B $P = 2.9 \times 10^{-1}$, $t_5 = -1.16$, paired t -test, $n = 6$ rats, Supplementary Table 2).

To assess whether the observed differences in cluster separability between light and dark conditions could arise by chance, we performed a permutation test. Within each rat, the darkness/light labels were randomly shuffled across trials while preserving the number of trials per condition. For each permutation, silhouette scores were recalculated from the layer embeddings of the trained network. This procedure was repeated 1000 times to generate an empirical null distribution of cluster separability. The observed silhouette scores for the dark condition consistently exceeded the 95th percentile of the null distribution, confirming that the enhanced separability in darkness reflects systematic differences in neural representations rather than random variation (Supplementary Figure 4).

To elucidate which electrodes held the most information about the floor textures, we performed occlusion analysis. The results revealed that electrodes in the forelimb subregion contributed more to successful texture and lighting predictions (Figure 6D, E, $P = 4.53 \times 10^{-6}$, $t_8 = -5.57$, Student's t -test, $n = 9$ rats), which aligns with our observation that negative deflections in LFPs were largest in forelimb-targeting channels (Figure 2E). Further analysis using class activation maps from integrated gradients showed spatiotemporal patterns of salient features unique to each experimental condition (Figure 6F). In particular, the average attribution score in the forelimb subregion displayed a temporal shift in the dark condition, implying an extended processing window for texture information when visual cues are unavailable (Figure 6G, Supplementary Figure 5).

Collectively, these results indicate that visual deprivation modifies population-level activity in S1, yielding more distinctive representations of tactile stimuli. Such reorganization could provide a neural substrate for enhanced tactile perception under conditions of reduced or absent visual input.

Discussion

This study provides evidence that S1 undergoes rapid reorganization of tactile representations when visual input is removed, even over a short period. Using high-density LFP recordings and deep learning techniques, we show that the neural encoding of tactile stimuli becomes more distinct under conditions of visual deprivation. Specifically, when rats were exposed to darkness, texture representations in S1 were more clearly distinguishable than under normal visual conditions. This demonstrates the adaptability of S1 and its capacity to rapidly adjust to changing sensory contexts.

A key finding was the significant increase in silhouette scores when texture representations were decoded in the dark. Silhouette scores, which quantify the separability of neural features, were notably higher under dark conditions, suggesting that visual deprivation sharpens the distinctions between tactile representations. While motivational factors, such as water restriction, could potentially confound the results, trial order reversals showed only a slight increase in silhouette scores during the dark condition. If motivation alone were responsible for the enhanced discriminability, silhouette scores should have been elevated during the later light trial in the reversed sequence. This supports the conclusion that the reorganization in S1 is more likely a

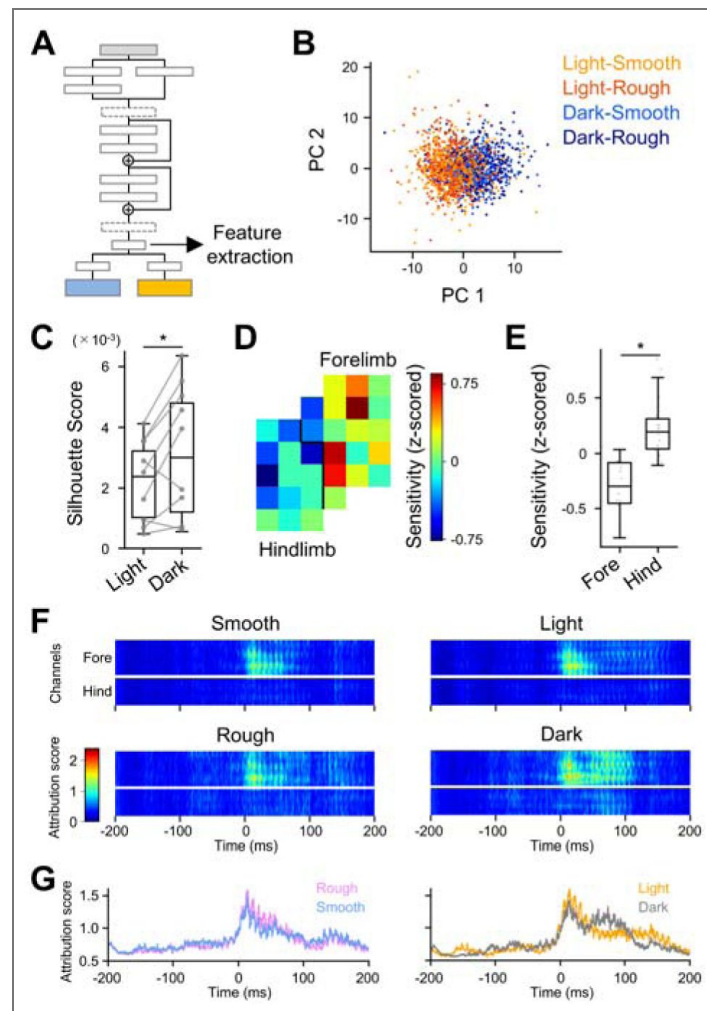


Figure 6. Neural representations are more distinct in dark environments than in light environments

(A) A 912-dimensional feature vector is extracted from the layer preceding the final output. (B) A scatter plot of these features from one rat shows individual LFP segments (aligned to forelimb contact). Light orange and dark orange correspond to the light conditions (smooth, rough), while light blue and dark blue correspond to the dark conditions (smooth, rough). (C) Silhouette scores across all nine rats, showing that the dark condition yields higher scores and thus more distinct neural representations. $P = 2.31 \times 10^{-2}$, $t_8 = -2.8$, paired t -test, $n = 9$ rats. (D) A pseudo-color map based on occlusion analysis, illustrating the contribution of each electrode in the forelimb and hindlimb subregions. Hotter regions indicate higher importance for the model's predictions. (E) Forelimb channels exhibit higher occlusion sensitivity than hindlimb channels, highlighting the forelimb's dominant role when the foot contacts the floor. $P = 4.53 \times 10^{-6}$, $t_{16} = -5.57$, Student's t -test, $n = 9$ rats. (F) Class activation maps generated via integrated gradients highlight key input features responsible for accurate model predictions of texture and environmental conditions. Attribution scores show each feature's impact on the model's output relative to a reference baseline: high positive scores denote features that strongly affect the predicted class. The onset of forelimb contact is aligned to time zero. (G) Attribution scores averaged over forelimb electrodes for floor texture (left) and the environmental conditions (right). A temporal lag in the dark condition suggests an extended processing window for floor texture when visual cues are absent.

direct result of visual deprivation rather than solely motivation or arousal. Although differences in silhouette scores were not significant in the reversed trials, this observation may reflect a lingering effect of prior visual deprivation, consistent with previous studies in humans showing that even brief periods of visual deprivation, such as blindfolding, can enhance tactile sensitivity. These findings suggest a robust and lasting effect of visual deprivation on tactile processing.

Our results also highlight the somatotopic specificity of tactile encoding in S1. The evoked potential in the forelimb subregion of S1 was significantly larger than in the hindlimb subregion, reflecting the topographic organization of S1, where distinct cortical areas process sensory inputs from different body parts (Ewert et al., 2008 [↗](#); Prsa et al., 2019 [↗](#); Sur et al., 1980 [↗](#)). Further, occlusion analysis revealed that forelimb subregions were critical for distinguishing between tactile textures and lighting conditions. This finding underscores the dominance of the forelimb subregion in encoding tactile information, especially during the specific task of walking and contacting different textures. Integrated gradient analysis confirmed the importance of forelimb channels in encoding texture information and showed that the temporal window of processing following forelimb contact was extended in the dark compared to the light condition. Additionally, comparative analysis of the integrated gradient and time-frequency maps revealed a negative correlation in the 50-60 Hz frequency range, while a positive correlation was observed in the 80-90 Hz range. These findings suggest that frequency-specific patterns of LFP activity in different conditions are closely linked to the texture representations captured by the CNN model.

The extended temporal window in the dark condition suggests that visual deprivation may enhance the retention of tactile information in S1. Previous studies have shown that S1 neurons are involved in the short-term retention of tactile information (Zhou and Fuster, 2000 [↗](#), 1997 [↗](#), 1996 [↗](#)), and prolonged neuronal firing in higher sensory areas may contribute to this sustained activity (Esmaili and Diamond, 2019 [↗](#); Leavitt et al., 2017 [↗](#)). Our findings suggest that the dark condition could lead to more prolonged neural representations of tactile stimuli. Whether this reflects a compensatory mechanism for the absence of visual input or a correlation with heightened tactile sensitivity requires further exploration.

Environmental illumination can also influence the arousal state, which in turn affects cortical dynamics as reflected in LFPs. Dark environments typically promote exploratory behavior and increase arousal, modulating cortical activity through irradiance- and cone-opponent-dependent pathways to arousal circuits like the locus coeruleus and basal forebrain (Tamayo et al., 2023 [↗](#)). Arousal in S1 enhances both baseline activity and stimulus encoding, with neuromodulatory drive sharpening temporal precision (Eggermann et al., 2014 [↗](#); Poulet and Petersen, 2008 [↗](#); Shimaoka et al., 2018 [↗](#)). These state-dependent effects may account for the enhanced separability of tactile representations observed in the dark, despite minimal differences in average evoked amplitude (Lee et al., 2020 [↗](#); McGinley et al., 2015 [↗](#); Shimaoka et al., 2018 [↗](#); Vinck et al., 2015 [↗](#)). The shift toward a desynchronized, high-arousal state in darkness likely reduces low-frequency shared variability and boosts fast-timescale signal components, improving the separability of neural clusters in a high-dimensional LFP space. Moreover, neuromodulatory engagement during heightened arousal can extend effective integration windows and enhance gain in task-relevant networks, consistent with the longer temporal window for accurate predictions seen in forelimb channels.

These findings highlight the limitations of traditional analysis methods, such as amplitude-based metrics and event-related potentials, which often fail to capture subtle, higher-dimensional features in neural signals (Saxena and Cunningham, 2019 [↗](#); D. L. Yamins and DiCarlo, 2016 [↗](#); D. L. K. Yamins and DiCarlo, 2016 [↗](#)). In contrast, our deep-learning approach revealed fine-grained spatiotemporal patterns in LFPs, showcasing enhanced texture-specific separability under visual deprivation. This demonstrates the power of advanced computational tools to uncover previously inaccessible shifts in sensory coding, with broad applicability to other high-dimensional neural datasets, such as those from multi-electrode arrays in freely behaving animals. However, it's important to note that each model was trained specifically for each rat, and inter-animal generalization remains a challenge, due to differences in electrode placement and individual brain

structure. Despite this limitation, the model's ability to extract complex, high-dimensional patterns from the dataset remains evident, and the results show significant progress in detecting sensory changes.

While fMRI studies have demonstrated tactile enhancement and corresponding neural reorganization during short-term visual deprivation (Facchini and Aglioti, 2003 [↗](#); Kauffman et al., 2002 [↗](#); Pascual-Leone and Hamilton, 2001b [↗](#)), the low temporal and spatial resolution of fMRI limits its ability to capture detailed changes in neural representations. Our study, through LFP recordings, provides a more nuanced picture of the population-level activity underlying tactile perception enhancement in the absence of visual input. Future studies could extend this work to assess whether rats can discriminate textures more accurately in the dark and explore the direct relationship between neural coding and perceptual performance.

From an evolutionary perspective, enhanced tactile sensitivity in low-visibility environments provides an adaptive advantage. Many species, including rodents, rely on somatosensation for navigation and foraging when visual information is scarce. The ability to rapidly enhance tactile processing in such conditions could aid in efficient resource acquisition and predator detection. Given the influence of arousal on tactile sensitivity (Lee et al., 2020 [↗](#); Shimaoka et al., 2018 [↗](#)), the dynamic nature of sensory processing becomes evident—S1 does not merely respond passively to tactile input but actively adapts its encoding strategies based on available sensory cues.

Several limitations warrant further exploration. First, if arousal contributes to the effects observed in S1 during light–dark transitions, we predict that arousal indices, such as pupil or whisking activity, should correlate with LFP spectral composition and decoding accuracy on a trial-by-trial basis (McGinley et al., 2015 [↗](#); Reimer et al., 2014 [↗](#); Shimaoka et al., 2018 [↗](#)). Second, manipulating neuromodulatory tone through optogenetic activation of arousal circuits should modulate the separability of texture representations, independent of illumination (Eggermann et al., 2014 [↗](#); Harris and Thiele, 2011 [↗](#)). Future experiments integrating real-time arousal measures and LFP analyses will be crucial in disentangling the contributions of arousal versus light–dark effects on cross-modal reorganization in S1. Additionally, while rats serve as a powerful model for sensory processing, their neural architecture may not fully capture the complexities of human perception. Extending this paradigm to assess perceptual changes in tactile acuity will further clarify the relationship between neural coding and perceptual performance.

In conclusion, our study provides new insights into the brain's remarkable ability to reorganize its sensory processing in response to changes in sensory input. By demonstrating that visual deprivation rapidly reconfigures tactile processing in S1, we highlight the brain's flexibility in adapting to environmental changes. This work underscores the potential of combining behavioral paradigms, LFP recordings, and deep learning techniques to understand the dynamic and adaptive nature of sensory coding, paving the way for future research in multisensory interactions.

Data availability

The data that support the findings of this study are available from the corresponding author upon request. Custom code generated during this study for data analysis are available at <https://github.com/UT-yakusaku/Yamashiro-eLife-2024> [↗](#).

Acknowledgements

This work was supported by JST ERATO (JPMJER1801), AMED-CREST (24wm0625401h0001; 24wm0625502s0501; 24wm0625207s0101; 24gm1510002s0104), the Institute for AI and Beyond of the University of Tokyo, JSPS Grants-in-Aid for Scientific Research (18H05525, 20K15926, 22K21353, 22J22097), KOSÉ Cosmetology Research Foundation, the Public Foundation of Chubu Science and Technology Center, and Konica Minolta Science and Technology Foundation.

Additional files

[Supplementary Materials](#) [↗](#)

Additional information

Funding

Funder	Grant reference number	Author
MEXT Japan Science and Technology Agency (JST)	https://doi.org/10.52926/jpmjer1801	Yuji Ikegaya
Japan Agency for Medical Research and Development (AMED)	24wm0625401h0001	Yuji Ikegaya
Japan Agency for Medical Research and Development (AMED)	24wm0625502s0501	Yuji Ikegaya
Japan Agency for Medical Research and Development (AMED)	24wm0625207s0101	Yuji Ikegaya
Japan Agency for Medical Research and Development (AMED)	24gm1510002s0104	Yuji Ikegaya
The Institute for AI and Beyond of the University of Tokyo		Yuji Ikegaya
MEXT Japan Society for the Promotion of Science (JSPS)	18H05525	Yuji Ikegaya
MEXT Japan Society for the Promotion of Science (JSPS)	20K15926	Nobuyoshi Matsumoto
MEXT Japan Society for the Promotion of Science (JSPS)	22K21353	Yuji Ikegaya
Cosmetology Research Foundation		Nobuyoshi Matsumoto
The Public Foundation of Chubu Science and Technology Center		Nobuyoshi Matsumoto
Konica Minolta Imaging Science Foundation (Konica Minolta Science and Technology Foundation)		Nobuyoshi Matsumoto
MEXT Japan Society for the Promotion of Science (JSPS)	22KJ1103	Kotaro Yamashiro

Author ORCID iDs

Kotaro Yamashiro: <https://orcid.org/0000-0002-8956-8529>

Nobuyoshi Matsumoto: <https://orcid.org/0000-0002-0426-9416>

Yuji Ikegaya: <https://orcid.org/0000-0002-6434-9469>

References

- Abraira VE, Ginty DD** (2013) The sensory neurons of touch. *Neuron* **79**:618-639
<https://doi.org/10.1016/j.neuron.2013.07.051> | PubMed
- Bensmaia SJ, Denchev PV, Dammann JF, Craig JC, Hsiao SS.** (2008) The representation of stimulus orientation in the early stages of somatosensory processing. *J Neurosci* **28**:776-786
<https://doi.org/10.1523/jneurosci.4162-07.2008> | PubMed
- Bola Ł, Siuda-Krzywicka K, Paplińska M, Sumera E, Zimmermann M, Jednoróg K, Marchewka A, Szwed M** (2017) Structural reorganization of the early visual cortex following Braille training in sighted adults. *Sci Rep* **7**:17448 <https://doi.org/10.1038/s41598-017-17738-8> | PubMed
- Boroojerdi B, Bushara KO, Corwell B, Immisch I, Battaglia F, Muellbacher W, Cohen LG** (2000) Enhanced excitability of the human visual cortex induced by short-term light deprivation. *Cereb Cortex* **10**:529-534 <https://doi.org/10.1093/cercor/10.5.529> | PubMed

- Bulusu V, Lazar L** (2024) Crossmodal associations between naturally occurring tactile and sound textures. *Perception* **53**:219-239 <https://doi.org/10.1177/03010066231224557> | PubMed
- Burton H** (2003) Visual cortex activity in early and late blind people. *J Neurosci* **23**:4005-4011 <https://doi.org/10.1523/jneurosci.23-10-04005.2003> | PubMed
- Bushnell MC, Duncan GH, Hofbauer RK, Ha B, Chen JI, Carrier B** (1999) Pain perception: is there a role for primary somatosensory cortex?. *Proc Natl Acad Sci U S A* **96**:7705-7709 <https://doi.org/10.1073/pnas.96.14.7705> | PubMed
- Butler JS, Foxe JJ, Fiebelkorn IC, Mercier MR, Molholm S** (2012) Multisensory representation of frequency across audition and touch: high density electrical mapping reveals early sensory-perceptual coupling. *J Neurosci* **32**:15338-15344 <https://doi.org/10.1523/jneurosci.1796-12.2012> | PubMed
- Delhaye BP, Long KH, Bensmaia SJ** (2018) Neural basis of touch and proprioception in primate cortex. *Compr Physiol* **8**:1575-1602 <https://doi.org/10.1002/cphy.c170033> | PubMed
- Dijkerman HC, de Haan EHF** (2007) Somatosensory processing subserving perception and action: Dissociations, interactions, and integration. *Behav Brain Sci* **30**:224-230 <https://doi.org/10.1017/s0140525x07001641>
- Di Plinio S, Perrucci MG, Aleman A, Ebisch SJH.** (2020) I am Me: Brain systems integrate and segregate to establish a multidimensional sense of self. *Neuroimage* **205**:116284 <https://doi.org/10.1016/j.neuroimage.2019.116284> | PubMed
- Driver J, Spence C** (2000) Multisensory perception: beyond modularity and convergence. *Curr Biol* **10**:R731-5 [https://doi.org/10.1016/s0960-9822\(00\)00740-5](https://doi.org/10.1016/s0960-9822(00)00740-5) | PubMed
- Eggermann E, Kremer Y, Crochet S, Petersen CCH** (2014) Cholinergic signals in mouse barrel cortex during active whisker sensing. *Cell Rep* **9**:1654-1660 <https://doi.org/10.1016/j.celrep.2014.11.005> | PubMed
- Eimer M, Forster B** (2003) Modulations of early somatosensory ERP components by transient and sustained spatial attention. *Exp Brain Res* **151**:24-31 <https://doi.org/10.1007/s00221-003-1437-1> | PubMed
- Esmaili V, Diamond ME** (2019) Neuronal correlates of tactile working memory in prefrontal and vibrissa somatosensory cortex. *Cell Rep* **27**:3167-3181. <https://doi.org/10.1016/j.celrep.2019.05.034> | PubMed
- Ewert TAS, Vahle-Hinz C, Engel AK** (2008) High-frequency whisker vibration is encoded by phase-locked responses of neurons in the rat's barrel cortex. *J Neurosci* **28**:5359-5368 <https://doi.org/10.1523/jneurosci.0089-08.2008> | PubMed
- Facchini S, Aglioti SM** (2003) Short term light deprivation increases tactile spatial acuity in humans. *Neurology* **60**:1998-1999 <https://doi.org/10.1212/01.wnl.0000068026.15208.d0> | PubMed
- Goldreich D, Kanics IM** (2003) Tactile acuity is enhanced in blindness. *J Neurosci* **23**:3439-3445 <https://doi.org/10.1523/jneurosci.23-08-03439.2003> | PubMed
- Goodwin AW, Wheat HE** (2004) Sensory signals in neural populations underlying tactile perception and manipulation. *Annu Rev Neurosci* **27**:53-77 <https://doi.org/10.1146/annurev.neuro.26.041002.131032> | PubMed
- Harris KD, Thiele A** (2011) Cortical State and Attention. *Nature Reviews Neuroscience* **12**:509 <https://doi.org/10.1038/nrn3084> | PubMed
- He K, Zhang X, Ren S, Sun J** (2015) Deep residual learning for image recognition. *arXiv* <https://doi.org/10.48550/arxiv.1512.03385>
- Hopkins K, Kass SJ, Blalock LD, Brill JC** (2017) Effectiveness of auditory and tactile crossmodal cues in a dual-task visual and auditory scenario. *Ergonomics* **60**:692-700 <https://doi.org/10.1080/00140139.2016.1198495> | PubMed

- Karlen SJ, Kahn DM, Krubitzer L (2006) Early blindness results in abnormal corticocortical and thalamocortical connections. *Neuroscience* **142**:843-858 <https://doi.org/10.1016/j.neuroscience.2006.06.055> | PubMed
- Kauffman T, Théoret H, Pascual-Leone A (2002) Braille character discrimination in blindfolded human subjects. *Neuroreport* **13**:571-574 <https://doi.org/10.1097/00001756-200204160-00007> | PubMed
- Koch KW, Fuster JM (1989) Unit activity in monkey parietal cortex related to haptic perception and temporary memory. *Exp Brain Res* **76**:292-306 <https://doi.org/10.1007/bf00247889> | PubMed
- Leavitt ML, Mendoza-Halliday D, Martinez-Trujillo JC (2017) Sustained activity encoding working memories: Not fully distributed. *Trends Neurosci* **40**:328-346 <https://doi.org/10.1016/j.tins.2017.04.004> | PubMed
- Lee CCY, Kheradpezhoh E, Diamond ME, Arabzadeh E (2020) State-dependent changes in perception and coding in the mouse somatosensory cortex. *Cell Rep* **32**:108197 <https://doi.org/10.1016/j.celrep.2020.108197> | PubMed
- Luna R, Hernández A, Brody CD, Romo R (2005) Neural codes for perceptual discrimination in primary somatosensory cortex. *Nat Neurosci* **8**:1210-1219 <https://doi.org/10.1038/nn1513> | PubMed
- McGinley MJ, David SV, McCormick DA (2015) Cortical Membrane Potential Signature of Optimal States for Sensory Signal Detection. *Neuron* **87**:179-192 <https://doi.org/10.1016/j.neuron.2015.05.038> | PubMed
- Merabet LB, Hamilton R, Schlaug G, Swisher JD, Kiriakopoulos ET, Pitskel NB, Kauffman T, Pascual-Leone A (2008) Rapid and reversible recruitment of early visual cortex for touch. *PLoS One* **3**:e3046 <https://doi.org/10.1371/journal.pone.0003046> | PubMed
- Moulton EA, Pendse G, Becerra LR, Borsook D (2012) BOLD responses in somatosensory cortices better reflect heat sensation than pain. *J Neurosci* **32**:6024-6031 <https://doi.org/10.1523/jneurosci.0006-12.2012> | PubMed
- Nikbakht N, Tafreshiha A, Zoccolan D, Diamond ME (2018) Supralinear and Supramodal Integration of Visual and Tactile Signals in Rats: Psychophysics and Neuronal Mechanisms. *Neuron* **97**:626-639. <https://doi.org/10.1016/j.neuron.2018.01.003> | PubMed
- Norman JF, Bartholomew AN (2011) Blindness enhances tactile acuity and haptic 3-D shape discrimination. *Atten Percept Psychophys* **73**:2323-2331 <https://doi.org/10.3758/s13414-011-0160-4> | PubMed
- Pascual-Leone A, Hamilton R (2001a) The metamodal organization of the brain. *Prog Brain Res* **134**:427-445 [https://doi.org/10.1016/s0079-6123\(01\)34028-1](https://doi.org/10.1016/s0079-6123(01)34028-1) | PubMed
- Pascual-Leone A, Hamilton R (2001b) The metamodal organization of the brain. *Prog Brain Res* **134**:427-445 [https://doi.org/10.1016/s0079-6123\(01\)34028-1](https://doi.org/10.1016/s0079-6123(01)34028-1) | PubMed
- Piras F, Vecchio D, Ciullo V, Gili T, Banaj N, Piras F, Spalletta G (2020) Sense of external agency is sustained by multisensory functional integration in the somatosensory cortex. *Hum Brain Mapp* **41**:4024-4040 <https://doi.org/10.1002/hbm.25107> | PubMed
- Poulet JFA, Petersen CCH (2008) Internal brain state regulates membrane potential synchrony in barrel cortex of behaving mice. *Nature* **454**:881-885 <https://doi.org/10.1038/nature07150> | PubMed
- Prsa M, Morandell K, Cuenu G, Huber D (2019) Feature-selective encoding of substrate vibrations in the forelimb somatosensory cortex. *Nature* **567**:384-388 <https://doi.org/10.1038/s41586-019-1015-8> | PubMed
- Reimer J, Froudarakis E, Cadwell CR, Yatsenko D, Denfield GH, Tolia AS (2014) Pupil fluctuations track fast switching of cortical states during quiet wakefulness. *Neuron* **84**:355-362 <https://doi.org/10.1016/j.neuron.2014.09.033> | PubMed
- Sadato N, Pascual-Leone A, Grafman J, Ibañez V, Deiber MP, Dold G, Hallett M (1996) Activation of the primary visual cortex by Braille reading in blind subjects. *Nature* **380**:526-528 <https://doi.org/10.1038/380526a0> | PubMed

- Salinas E, Hernandez A, Zainos A, Romo R (2000) Periodicity and firing rate as candidate neural codes for the frequency of vibrotactile stimuli. *J Neurosci* **20**:5503-5515 <https://doi.org/10.1523/jneurosci.20-14-05503.2000> | PubMed
- Saxena S, Cunningham JP (2019) Towards the neural population doctrine. *Curr Opin Neurobiol* **55**:103-111 <https://doi.org/10.1016/j.conb.2019.02.002> | PubMed
- Schürmann M, Caetano G, Jousmäki V, Hari R (2004) Hands help hearing: facilitatory audiotactile interaction at low sound-intensity levels. *J Acoust Soc Am* **115**:830-832 <https://doi.org/10.1121/1.1639909> | PubMed
- Serino A (2019) Peripersonal space (PPS) as a multisensory interface between the individual and the environment, defining the space of the self. *Neurosci Biobehav Rev* **99**:138-159 <https://doi.org/10.1016/j.neubiorev.2019.01.016> | PubMed
- Shimaoka D, Harris KD, Carandini M (2018) Effects of arousal on mouse sensory cortex depend on modality. *Cell Rep* **22**:3160-3167 <https://doi.org/10.1016/j.celrep.2018.02.092> | PubMed
- Sugiyama S, Kinukawa T, Takeuchi N, Nishihara M, Shioiri T, Inui K (2019) Tactile Cross-Modal Acceleration Effects on Auditory Steady-State Response. *Front Integr Neurosci* **13**:496317 <https://doi.org/10.3389/fnint.2019.00072> | PubMed
- Sundararajan M, Taly A, Yan Q (2017) Axiomatic attribution for deep networks. *arXiv* <https://doi.org/10.48550/arxiv.1703.01365>
- Sur M, Merzenich MM, Kaas JH. (1980) Magnification, receptive-field area, and “hypercolumn” size in areas 3b and 1 of somatosensory cortex in owl monkeys. *J Neurophysiol* **44**:295-311 <https://doi.org/10.1152/jn.1980.44.2.295> | PubMed
- Tamayo E, Moulund JW, Lucas RJ, Brown TM (2023) Regulation of mouse exploratory behaviour by irradiance and cone-opponent signals. *BMC Biol* **21**:178 <https://doi.org/10.1186/s12915-023-01663-6> | PubMed
- Van Boven RW, Hamilton RH, Kauffman T, Keenan JP, Pascual-Leone A. (2000) Tactile spatial resolution in blind braille readers. *Neurology* **54**:2230-2236 <https://doi.org/10.1212/wnl.54.12.2230>
- Vinck M, Batista-Brito R, Knoblich U, Cardin JA (2015) Arousal and locomotion make distinct contributions to cortical activity patterns and visual encoding. *Neuron* **86**:740-754 <https://doi.org/10.1016/j.neuron.2015.03.028> | PubMed
- Wong M, Gnanakumaran V, Goldreich D (2011) Tactile spatial acuity enhancement in blindness: evidence for experience-dependent mechanisms. *J Neurosci* **31**:7028-7037 <https://doi.org/10.1523/jneurosci.6461-10.2011> | PubMed
- Yamashiro K, Aoki M, Matsumoto N, Ikegaya Y (2020) Polyherbal formulation enhancing cerebral slow waves in sleeping rats. *Biol Pharm Bull* **43**:1356-1360 <https://doi.org/10.1248/bpb.b20-00285> | PubMed
- Yamashiro K, Ikegaya Y, Matsumoto N (2024) Automatic detection of foot-strike onsets in a rhythmic forelimb movement. *Neurosci Res* **206**:41-50 <https://doi.org/10.1016/j.neures.2024.04.002> | PubMed
- Yamins DL, DiCarlo JJ (2016) Eight open questions in the computational modeling of higher sensory cortex. *Curr Opin Neurobiol* **37**:114-120 <https://doi.org/10.1016/j.conb.2016.02.001> | PubMed
- Yamins DLK, DiCarlo JJ (2016) Using goal-driven deep learning models to understand sensory cortex. *Nat Neurosci* **19**:356-365 <https://doi.org/10.1038/nn.4244> | PubMed
- Zeiler MD, Fergus R. (2013) Visualizing and understanding convolutional networks. *arXiv* <https://doi.org/10.48550/arxiv.1311.2901>
- Zhou YD, Fuster JM (2000) Visuo-tactile cross-modal associations in cortical somatosensory cells. *Proc Natl Acad Sci U S A* **97**:9777-9782 <https://doi.org/10.1073/pnas.97.17.9777> | PubMed
- Zhou YD, Fuster JM (1997) Neuronal activity of somatosensory cortex in a cross-modal (visuo-haptic) memory task. *Exp Brain Res* **116**:551-555 <https://doi.org/10.1007/pl00005783> | PubMed

Zhou YD, Fuster JM (1996) Mnemonic neuronal activity in somatosensory cortex. *Proc Natl Acad Sci U S A* **93**:10533-10537 <https://doi.org/10.1073/pnas.93.19.10533> | PubMed

Ziegler K, Folkard R, Gonzalez AJ, Burghardt J, Antharvedi-Goda S, Martin-Cortecero J, Isaías-Camacho E, Kaushalya S, Tan LL, Kuner T, *et al.* (2023) Primary somatosensory cortex bidirectionally modulates sensory gain and nociceptive behavior in a layer-specific manner. *Nat Commun* **14**:2999 <https://doi.org/10.1038/s41467-023-38798-7> | PubMed

Peer reviews

Reviewer #1 (Public review):

Summary:

The authors aimed to investigate how short-term visual deprivation influences tactile processing in the primary somatosensory cortex (S1) of sighted rats. They justify the study based on previous studies that have shown that long-term blindness can enhance tactile perception, and aim to investigate the change in neural representations underlying rapid, short-term cross-modal effects. The authors recorded local field potentials from S1 as rats encountered different tactile textures (smooth and rough sandpaper) under light and dark conditions. They used deep learning techniques to decode the neural signals and assess how tactile representations changed across the four different conditions. Their goal was to uncover whether the absence of visual cues leads to a rapid reorganization of tactile encoding in the brain.

Strengths:

The study effectively integrates high-density local field potential (LFP) recordings with convolutional neural network (CNN) analysis. This combination allows for decoding high-dimensional population-level signals, revealing changes in neural representations that traditional analyses (e.g., amplitude measures) failed to detect. The custom treadmill paradigm permits independent manipulation of visual and tactile inputs under stable locomotion conditions. Gait analysis confirms that motor behavior was consistent across conditions, strengthening the conclusion that neural changes are due to sensory input rather than movement artifacts.

Weaknesses:

- (1) While the study interprets the emergence of more distinct texture representations in the dark as evidence of rapid cross-modal plasticity, the claim rests on correlational data from a short-term manipulation and decoding analysis. The authors show that CNN-derived feature embeddings cluster more clearly by texture in the dark, but this does not directly demonstrate plasticity in the classical sense (e.g., synaptic or circuit-level reorganization). The authors have noted this as a limitation and have clarified that the observed changes reflect functional reorganization rather than structural plasticity.
- (2) Although gait was controlled, changes in arousal or exploratory behavior in light versus dark conditions might play a role in the observed neural differences. The authors have controlled for various factors in relation to locomotion, but future studies would benefit from more direct behavioural readouts of arousal states (e.g., via pupillometry or cortical state indicators).
- (3) It should be noted that the time course of the observed changes (within 10 minutes) is quite rapid, and while intriguing, the study does not include direct evidence that the underlying circuits were reorganized-only that population-level signals become more discriminable. The authors have adequately discussed this as an avenue for more mechanistic future research.

(4) The authors have adequately discussed that, while these findings are consistent with somatotopy and context-dependent dynamics, they do not provide strong independent evidence for novel spatial or temporal organization.

(5) The authors have also discussed that, while the neural data suggest enhanced tactile representations, the study does not assess whether rats' actual tactile perception improved. Future studies including an assessment of a behavioral readout (e.g., discrimination accuracy), would be insightful.

(6) The authors' discussion about the implications for sensory rehabilitation, including Braille training and haptic feedback enhancement was a bit premature, but they have amended this, and it remains an interesting translational potential to be explored in future studies.

(7) While the CNN showed good performance, more transparent models (e.g., linear classifiers or dimensionality reduction) appear to not exceed chance level. The implications of this are that there is an underlying complex structure in the LFPs that has yet to be fully uncovered, on the mechanistic level. This would be important to push the findings forward in future studies.

Therefore, while the authors raise interesting hypotheses around rapid plasticity, somatotopic dynamics, and rehabilitation, the evidence for each is indirect. Stronger claims will require future causal experiments, behavioral readouts, and mechanistic specificity beyond what the current data provides. However, the work represents an interesting starting point to a more mechanistic understanding in the future.

<https://doi.org/10.7554/eLife.106554.2.sa2>

Reviewer #2 (Public review):

Summary:

Yamashiro et al. investigated how transient absence of visual input (i.e. darkness) impacts tactile neural encoding in the rat primary somatosensory cortex (S1). They recorded local field potentials (LFPs) using a 32-channel array implanted in forelimb and hindlimb primary somatosensory cortex while rats walked on smooth or rough textures under illuminated and dark conditions. Employing a convolutional neural network (CNN), they successfully decoded both texture and lighting conditions from the LFPs. The authors conclude that the subtle differences in LFP patterns underlie tactile representation surface roughness and become more distinct in darkness, suggesting a rapid cross-modal reorganization of the neural code for this sensory feature.

Strengths:

- The manuscript addresses a valuable question regarding how sensory cortices dynamically adapt to changes in sensory context.
- The use of machine learning (CNNs) enables the analysis to go beyond conventional amplitude-based metrics, potentially uncovering subtle but meaningful effects.
- The authors have substantially improved the manuscript with clearer figures, additional statistical analyses (including permutation tests and cross-validation), and greater methodological transparency.

Weaknesses:

- The new analyses (grand-average LFPs, correlation maps, wavelet decompositions, attribution-score correlations) improve transparency but do not yet clarify which specific neural features the CNN exploits, leaving the central interpretability question unresolved.
- A plausible alternative explanation for the increased discriminability in darkness remains insufficiently ruled out: visually driven activity in the light condition (e.g., ambient illumination changes or self-motion-induced visual input) could contaminate S1 LFPs and account for the effect without reflecting a true neural representational change.
- Behavioural and order controls have been improved but remain somewhat limited in sample size.

Overall assessment:

The revised manuscript is clearer, more transparent, and technically strengthened. However, the true nature of the signal changes underlying the observed differences in discriminability remains unclear, limiting the scientific strength of the conclusions. The possibility that visual interference contributes to the observed effects remains a plausible and untested alternative interpretation. Additional experiments or analyses quantifying visually evoked activity in S1 would be required to confirm the claim of genuine reorganization of neural representation depending on the illumination condition.

<https://doi.org/10.7554/eLife.106554.2.sa1>

Author Response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

(1) While the study interprets the emergence of more distinct texture representations in the dark as evidence of rapid cross-modal plasticity, the claim rests on correlational data from a short-term manipulation and decoding analysis. The authors show that CNN-derived feature embeddings cluster more clearly by texture in the dark, but this does not directly demonstrate plasticity in the classical sense (e.g., synaptic or circuit-level reorganization).

Thank you for this insightful comment. We acknowledge that our claim of “rapid cross-modal plasticity” is based on correlational evidence and does not directly address synaptic or circuit-level reorganization, which would require more invasive methods. Our study instead focuses on changes in the representational structure of tactile stimuli when visual input is temporarily removed, highlighting the adaptability of sensory coding to environmental context. We agree that this distinction is important and have revised the manuscript to clarify that the observed changes reflect functional reorganization rather than structural plasticity, as indicated by the enhanced separability of texture representations in S1 during darkness.

(2) Although gait was controlled, changes in arousal or exploratory behavior in light versus dark conditions might contribute to the observed neural differences. These factors are acknowledged but not directly measured (e.g., via pupillometry or cortical state indicators).

Thank you for your insightful comment. We agree that arousal and exploratory behavior could influence neural differences and have considered these factors in our study. While gait was controlled, we did not directly measure arousal (e.g., via pupillometry or cortical indicators).

To partially address this, we reviewed locomotor-speed traces (Supplementary Figure 1), which showed no significant differences between light and dark conditions, suggesting movement speed did not drive the neural differences. We also reversed the order of light and dark conditions, and although the separability of textures was not significantly different, it further supports that motivation did not confound our results.

However, we acknowledge that arousal may still affect cortical dynamics, especially in the dark condition, where the lack of visual input might alter exploratory behavior. Due to technical limitations, we could not directly measure arousal states, and this is now discussed in the revised manuscript. While we cannot rule out the influence of arousal, the enhanced separability of texture representations suggests that sensory reorganization due to visual deprivation likely played a substantial role.

(3) Moreover, the time course of the observed changes (within 10 minutes) is quite rapid, and while intriguing, the study does not include direct evidence that the underlying circuits were reorganized - only that population-level signals become more discriminable. As such, the term "plasticity" may overstate the conclusions and should be interpreted with caution unless validated by additional causal or longitudinal data.

Thank you for your important comment. We agree that the term "plasticity" may overstate our conclusions, as our study focuses on population-level signal changes rather than direct evidence of circuit-level reorganization.

To address this, we have revised the manuscript to clarify that while the observed changes in neural separability suggest functional reorganization of sensory representations, they do not confirm structural plasticity. We have updated the wording throughout the manuscript to emphasize that these findings reflect functional reorganization in response to short-term visual input loss, rather than structural or long-term plasticity.

We also updated the discussion to highlight the need for future research with more invasive approaches to validate the causal mechanisms behind these rapid changes in neural dynamics.

(4) The study highlights the forelimb region of S1 and a post-contact temporal window as particularly important for decoding texture, based on occlusion and integrated gradient analyses. However, this finding may be somewhat circular: The LFPs were aligned to forelimb contact, and the floor textures were sensed primarily via the forelimbs, making it unsurprising that forelimb electrodes were most informative. The observed temporal window corresponds directly to the event-aligned epoch, and while it may shift slightly in duration in the dark, this could reflect general differences in sensory gain or arousal, rather than changes in stimulus-specific encoding. Thus, while these findings are consistent with somatotopy and context-dependent dynamics, they do not provide strong independent evidence for novel spatial or temporal organization.

Thank you for your insightful comment. We understand your concern that the finding of forelimb electrodes being most informative might seem circular, given that the LFPs were aligned to forelimb contact, and the floor textures were primarily sensed by the forelimbs. This design choice was intentional, as the task focused on texture perception through the forelimb, and the forelimb subregion of S1 is naturally expected to play a dominant role in this process. While this somatotopic specificity may make the results predictable, our aim

was to emphasize the changes in temporal dynamics of neural processing under visual deprivation.

We observed a shift in the temporal window's duration in the dark condition, which we interpret as a change in how texture information is processed without visual input. While this could reflect sensory gain or arousal differences, the lack of significant differences in locomotor speed or other behavioral measures (Supplementary Figure 1) suggests that these changes are more likely due to functional reorganization of sensory processing.

We have clarified in the discussion that the shift in the temporal window is consistent with previous research on sensory reorganization involving both spatial and temporal cortical adjustments. While we do not claim novel spatial or temporal organization, we emphasize that the shift in temporal dynamics suggests adaptation in encoding strategy for texture perception in the absence of visual input. Future studies measuring arousal states (e.g., pupil diameter or cortical state markers) would help distinguish the contributions of arousal versus sensory reorganization to these dynamics.

(5) While the neural data suggest enhanced tactile representations, the study does not assess whether rats' actual tactile perception improved. Without a behavioral readout (e.g., discrimination accuracy), claims about perceptual enhancement remain speculative.

Thank you for raising this important point. We agree that while the neural data suggest enhanced separability of tactile representations in the dark condition, we do not directly assess whether these changes translate into improved tactile perception behaviorally.

However, the primary aim of our study is not to claim perceptual enhancement, but to demonstrate that neural representations in the somatosensory cortex can rapidly reorganize in response to visual deprivation. To clarify this distinction, we have revised the manuscript to emphasize that the observed neural changes in S1 are consistent with functional reorganization of tactile representations, rather than a direct indication of perceptual improvement.

Future studies will be crucial to directly test whether the enhanced separability of tactile representations in S1 correlates with improved tactile perception in a behavioral task. We have highlighted this as an avenue for future research to better understand the link between neural changes and perceptual outcomes.

(6) In addition to point 4, the authors discuss implications for sensory rehabilitation, including Braille training and haptic feedback enhancement. However, the lack of actual chronic or even more acute pathological sensory deprivation, behavioral data, or subsequent intervention in this study limits the ability to draw translational conclusions. It remains unknown whether the more distinct neural representations observed actually translate into better tactile performance, discriminability, or perception. Additionally, extrapolating from rats walking on sandpaper in the dark to human rehabilitative contexts is speculative without a clearer behavioral or mechanistic bridge. The potential is certainly there, but the claim is currently aspirational rather than empirically grounded.

Thank you for raising this important point. Upon careful consideration, we have decided to remove the discussion of sensory rehabilitation implications from the revised manuscript. We have refocused the manuscript to concentrate solely on the neural findings related to tactile encoding reorganization in response to short-term sensory deprivation, avoiding speculative extrapolation to human rehabilitative contexts. This revised approach ensures that the manuscript emphasizes the empirical findings without overstating the translational potential.

(7) While the CNN showed good performance, details on generalization robustness and validation (e.g., cross-validation folds, variance across animals) are not deeply discussed. Also, while explainability tools were used, interpretability of CNNs remains limited, and more transparent models (e.g., linear classifiers or dimensionality reduction) could offer complementary insights.

We appreciate the reviewer's valuable feedback. In response to the concern about generalization robustness and validation, we have now conducted 5-fold cross-validation to assess the model's performance within animals (Figure 6C). We also have added supplementary information on the average silhouette scores across the different folds and animals (Supplementary Table 1, 2). These details are provided in the methods section and discussed in the results to offer a clearer picture of the model's robustness and consistency across rats.

Regarding the interpretability of CNNs, we acknowledge that deep learning models can lack transparency. We also attempted classification using more transparent models such as PCA and SVM, but their performance did not exceed chance level (Supplementary Figure 2). This indicates that while these simpler models are more interpretable, they cannot capture the complex representations in the LFPs, making deep learning models like CNNs necessary for extracting these insights.

Reviewer #2 (Public review):

(1) Despite applying explainability techniques to the CNN-based decoder, the study does not clearly demonstrate the precise "subtle, high-dimensional patterns" exploited by the CNN for surface roughness decoding, limiting the physiological interpretability of the results. Additional analyses (e.g., detailed waveform morphology analysis on grand averages, time-frequency decompositions, or further use of explainability methods) are necessary to clarify the exact nature of the discriminative activity features enabling the CNN to decode surface roughness and how these change with the sensory context (i.e., in light or darkness).

Thank you for your insightful comment. We recognize the importance of clarifying the exact nature of the high-dimensional neural patterns that the CNN exploits for surface roughness decoding. In response, we have performed additional analyses to provide a more detailed explanation of the CNN's decision-making process and the discriminative features it learned:

Grand-Average LFP Waveforms Analysis: We calculated the grand-average LFP waveforms for each texture × lighting condition (Figure 4A). While visual inspection did not reveal distinct features in the averaged waveforms, we explored the channel-wise correlations between textures under both light and dark conditions (Figure 4B). We found that the correlation between textures was lower in the dark condition, suggesting that LFPs become more distinct between textures when visual input is absent, which aligns with the CNN's output.

Time-Frequency Decomposition (Wavelet Analysis): We also performed time-frequency decomposition of the LFPs using wavelet transforms (Figure 4D). No prominent differences emerged across texture × lighting conditions in the spectral domain. However, upon computing differences in wavelet features between light and dark conditions and analyzing the relationship with the CNN's attribution scores (Supplementary Figures 5A-C), we observed a negative correlation in the 50-60 Hz range and a positive correlation in the 80-90 Hz range. This suggests frequency-specific modulation in LFP activity that may contribute to texture representations, providing further support for the CNN's learned features.

(2) The claim regarding cross-modal representation reorganization heavily relies on a silhouette analysis (Figure 5C), which shows a modest effect size and borderline

statistical significance ($p=0.05$ with $n=9+2$). More rigorous statistical quantification, such as permutation tests and reporting underlying cluster distances for all animals, would strengthen confidence in this finding.

Thank you for your thoughtful comment. We appreciate your suggestion to strengthen the statistical rigor of our analysis regarding the cross-modal representation reorganization. In response, we have implemented several additional analyses to more rigorously quantify the separability of neural representations between light and dark conditions:

(1) Permutation Test for Cluster Separability: We performed a permutation test to assess whether the observed differences in cluster separability between light and dark conditions were statistically significant or could have arisen by chance. The results showed that the silhouette scores for the dark condition consistently exceeded the 95th percentile of the null distribution (Supplementary Figure 4). This permutation test strengthens the validity of our findings, indicating that the enhanced separability in darkness is a systematic reorganization of neural representations, not due to random fluctuations.

(2) Reporting Cluster Distances: To address concerns about the modest effect size and borderline significance, we have explicitly reported the underlying cluster distances in the form of silhouette scores for each individual animal (Supplementary Table 1, 2). These values reflect the Euclidean distance between clusters within each rat, providing a clearer understanding of the separability observed.

(3) Additional Statistical Analysis on Silhouette Scores: To further enhance the rigor of our statistical analysis, we recalculated the silhouette scores using 5-fold cross-validation within each animal, ensuring that our results are robust across multiple data splits (Figure 6C).

By incorporating these additional analyses and reporting detailed cluster distances, we believe we have significantly strengthened the confidence in our claim of cross-modal reorganization.

(3) While the authors recorded in the somatosensory cortex, primarily known for its tactile responsivity, I would be cautious not to rule out a priori the presence of crossmodal (visual) responses in the area. In this case, the stronger texture separation in darkness might be explained by the absence of some visually-evoked potentials (VEPs) rather than genuine cross-modal reorganization. Clarification is needed to rule out visual interference and this would strengthen the claim.

Thank you for raising this important point. In response to your concern, we carefully examined whether visually-evoked potentials (VEPs) could be present in the S1 recordings, particularly under the light condition. However, we observed that this experiment did not involve any cue-guided visual stimulation, such as flashing lights or visual cues aligned with the LFP recordings. Without such external visual stimuli, it is unlikely that VEPs would be reliably evoked in the S1. Therefore, we believe the stronger texture separation observed in the dark condition is not due to visual interference, but rather reflects a genuine sensory reorganization in response to the absence of visual input.

(4) Behavioural controls are limited to gross gait parameters; more detailed analyses of locomotor behavior and additional metrics (e.g., pupil size or locomotor variance) would robustly rule out potential arousal or motor confounds.

Thank you for your insightful comment regarding behavioral controls. In response, we have added locomotor speed traces aligned with corresponding LFPs (Supplementary Figure 1) to demonstrate that locomotion remained consistent across trials, irrespective of environmental condition (light vs. dark). Additionally, we report locomotor speed variance over 10-minute blocks to confirm no significant motor changes affecting neural recordings. These analyses indicate that LFP differences are unlikely due to locomotor confounds.

While measuring pupil size could be useful for assessing arousal, the camera resolution in our study was insufficient for reliable measurements. We have noted this limitation in the Discussion and recommend that future studies with high-resolution eye-tracking explore arousal's role in sensory processing in S1.

(5) The consistent ordering of trials (10 minutes of light then 10 minutes of dark) could introduce confounds such as fatigue or satiation (and also related arousal state), which should be controlled by analyzing sessions with reversed condition ordering.

Thank you for highlighting the potential confounds due to trial ordering. To address this, we reversed the condition order (dark before light) in a subset of sessions from six rats and reanalyzed the data (Supplementary Figure 3). The results showed not significant, but increase separability in the dark condition, suggesting that the enhanced separability in the dark condition is not due to trial order effects like fatigue or satiation. While order effects may contribute to trial-to-trial variability, the consistent pattern of enhanced separability in the dark further supports the interpretation that visual deprivation directly influences the reorganization of tactile representations in S1.

(6) The focus on forelimb-aligned LFP analyses raises the possibility that hindlimb-aligned data might yield different conclusions, suggesting alignment effects might bias the results.

Thank you for your insightful comment on the potential bias of forelimb-aligned LFP analyses. We acknowledge that the choice of alignment event can influence the results and appreciate the suggestion to consider hindlimb-aligned data. However, our experimental design specifically focused on forelimb S1. The forelimb region of S1 was oversampled in our array, and as expected, we observed larger responses there, consistent with the known somatotopic organization of S1.

While hindlimb-aligned data could provide additional insights, it is not directly relevant to the primary question of how forelimb S1 codes tactile information under visual deprivation. We do not believe the forelimb alignment introduces a bias, as it aligns with the sensory task being investigated. However, we recognize the value of exploring alternative alignments and have now included a discussion in the Methods section regarding the rationale for our design choices.

(7) The authors' dismissal of amplitude-based metrics as ineffective is inadequately substantiated. A clearer demonstration (e.g., event-related waveforms averaged by conditions, presented both spatially and temporally) would support this claim.

Thank you for your constructive comment. In response, we have added a more detailed analysis of event-related waveforms, averaged across conditions (light vs. dark, smooth vs. rough textures), and presented them spatially and temporally aligned to forelimb contact (Figure 4A). These waveforms did not show clear, distinct features that could differentiate conditions, which highlights the limitations of traditional amplitude-based metrics in detecting subtle neural activity changes related to visual deprivation.

We further performed channel-wise correlation analyses (Figure 4B), revealing stronger texture correlations in the light condition, indicating that averaged waveforms do not capture the nuanced differences in neural dynamics. Additionally, time-frequency spectrograms and channel-channel correlation matrices (Figures 4C and 4D) did not show distinct condition differences, reinforcing the limitations of amplitude-based metrics.

These findings, along with the superior performance of machine learning-based decoding methods (e.g., CNN), support our claim that amplitude-based approaches are insufficient for fully capturing the complexity of the neural data.

(8) Wording ambiguity regarding "attribution score" versus "activation amplitude" (Figure 5) complicates the interpretation of key findings. This distinction must be clarified for proper assessment of the results.

Thank you for pointing out the ambiguity between "attribution score" and "activation amplitude." To address this, we have revised the manuscript to use "attribution score" only.

(9) Generalization across animals remains unaddressed. The current within-subject decoding setup limits conclusions regarding shared neural representations across individuals. Adopting cross-validation strategies and exploring between-animal analyses would add significant value to the manuscript.

Thank you for highlighting the importance of generalization across animals. While our study focused on within-subject decoding, we acknowledge that this limits conclusions about shared neural representations across individuals. We expect that inter-animal generalization would be challenging, as models trained on data from a single rat may not perform well on data from others due to differences in electrode placement, brain anatomy, and neural representations. We recognize the value of cross-validation strategies and between-animal analyses and will consider them in future work to address this limitation.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) I would strongly recommend that the authors refine their introduction to be more concise. Many concepts and study aims are repeated many times and, therefore, present as highly redundant text. The introduction may be half the length and still contain the important concepts to set up the justification for the study. I would also suggest refining to be less about sensory deprivation (e.g., with blindness) and more in relation to context, as the acute nature of the study allows one to conclude more about the latter than the former.

Thank you for your feedback on the introduction. We have revised the section to reduce redundancy and present the key concepts more concisely. We also streamlined the study aims and focused more on the context of the acute nature of the study, as you suggested, rather than emphasizing sensory deprivation. This revision better aligns with the main focus of the research and improves clarity. We believe the updated introduction provides a more direct justification for the study.

(2) I am not sure if Figures 1-3 are meant to be in grey-scale for some reason (perhaps to represent light and dark), but I would encourage the authors to examine if this is necessary, as the use of color generally helps one more easily follow Figures.

Thank you for this suggestion. Upon review, we agree that the use of color would enhance the clarity and readability of our figures. We have revised the figures including the newly added supplementary figures to incorporate color.

(3) Figure 5, Figure legend title - check wording.

Thank you for pointing this out. The title has been adjusted for consistency with the other figure legends.

Reviewer #2 (Recommendations for the authors):

(1) Analyses that would strengthen the main claims (major):

(a) Identify the features exploited by the CNN.

(i) Provide grand-average LFP waveforms for each texture × lighting condition (fore- and hind-limb channels shown separately, spatially arranged as in Figure 3C) and try to relate them to the decoding strategy learned by the CNN.

Thank you for your helpful suggestion. We have calculated the grand-average LFP waveforms for each texture × lighting condition and included them in Figure 4A, with fore- and hind-limb channels shown separately and spatially arranged as in Figure 3C. Upon visual inspection, the mean waveforms did not reveal clear, distinct features. To further investigate, we computed the channel-wise correlation between different textures under both dark and light conditions. By subtracting the correlation coefficients for the dark environment from those in the light, we observed that the correlation between textures was lower in the dark environment (Figure 4B). This suggests that LFPs are more distinct between textures in the dark, supporting the CNN model's output. However, this also indicates that the CNN has captured more complex, nuanced information, as it is able to discriminate between LFPs on a single-trial basis, rather than relying on mean traces.

To assess how the correlation between average LFP waveforms varied across channels, we also calculated the channel-channel correlation matrix for all 32 channels in each condition. While we found stronger correlations within each S1 subregion, we did not observe clear differences of correlation matrix between light and dark conditions, nor between different textures (Figure 4C).

(ii) Add channel-wise and time-frequency maps (e.g., wavelet or spectrograms) for each texture × lighting condition and try to relate them to the decoding strategy learned by the CNN.

Thank you for the valuable suggestion. We calculated wavelet features for each LFP segment and averaged them across trials to assess differences in LFP between light and dark conditions, as well as across textures (Figure 4D). However, no distinct differences were observed in the spectral map. To investigate further, we computed the differences in spectral maps for LFPs in light and dark trials. We then calculated the difference in attribution scores derived from the integrated gradient map (Supplementary Figure 4A). Subsequently, we calculated the correlation coefficients between the differences in integrated gradients and the differences in power across each frequency band in the spectral map (Supplementary Figures 4B and 4C). A negative correlation was found in the 50-60 Hz range, while a positive correlation was observed in the 80-90 Hz range. These findings suggest that frequency-specific patterns of LFP activity in different conditions may be linked to the texture representations captured by the CNN model. We have included a discussion of these findings in [lines 463-468].

(b) Quantify the "enhanced separability in darkness" more rigorously.

(i) Report cluster-distances (e.g. Euclidean) for each individual animal.

We thank the reviewer for this helpful comment. When calculating the silhouette score, we used Euclidean distance as the distance metric. The silhouette score is defined for each data point as the difference between the average distance to points within its assigned cluster and the average distance to points in the nearest other cluster, normalized by the larger of the two values. Thus, the silhouette score inherently reflects the relative cluster distances both within and across conditions for each individual animal. Because we report and statistically analyze silhouette scores (Figure 6C), these values already quantify and compare the Euclidean cluster distances across conditions at the animal level. For clarity, we have now added a definition of the silhouette score in the Methods section of the main text [lines 269-278]. We also included the calculated silhouette scores in Supplementary Table 1.

(ii) Run a permutation or bootstrap test (shuffling darkness/light labels within animals) to obtain an empirical null distribution for cluster separability in the network embedding space.

We thank the reviewer for this important suggestion. In response, we implemented a permutation test to assess the robustness of our cluster separability results. Specifically, we shuffled the darkness/light labels within each animal and recalculated silhouette scores across 1000 resamples to generate an empirical null distribution. The observed separability between light and dark conditions consistently exceeded the 95th percentile of the null distribution (Supplementary Figure 3). This confirms that the enhanced cluster separability in darkness was not attributable to random fluctuations in labeling but instead reflected a systematic reorganization of neural representations.

(c) Control for possible visually-evoked potentials (VEPs).

(i) Search the LFPs recorded in light for stereotyped VEP components and/or comment on this possible confound (i.e., VEPs in S1?).

Thank you for raising this point. Although it would be interesting to observe if a VEP is present in the S1 of rats, this experiment did not involve cue-guided visual stimulation. Additionally, there was no environmental visual cue that could serve as an external trigger to align the LFPs for VEP analysis in S1. Furthermore, since even the somatosensory evoked potential was not clearly visible in the S1 LFP without averaging the aligned LFPs, it is unlikely that we would be able to observe VEPs in single trials.

(d) Address behavioral and arousal confounds.

(i) Provide example locomotor-speed traces (aligned with corresponding LFPs) and report locomotor-speed variance across the 10-min blocks.

Thank you for your comment. We had speedometer installed for the recording of the last two rats. We have now provided example speed traces and the speed variance across blocks in Supplementary Figure 1. The traces show that the locomotor-speed was stable in each trial.

(ii) If available from the camera recordings, include pupil diameter as a proxy for arousal; otherwise, discuss explicitly how arousal changes might affect S1 LFPs.

Thank you for this suggestion. We strongly agree that measuring pupil diameters should be incorporated into future studies. However, because our camera did not have sufficient resolution to capture pupil diameters, we have addressed this limitation in the discussion section [lines 525-537].

(e) Address order effects (and motivation/satiety confounds)

(i) Present at least a subset of sessions in which the dark block precedes the light block; re-analyze the silhouette score/discriminability with block order as a factor.

Thank you for this helpful suggestion. We conducted additional analyses using sessions from 6 rats in which the dark block preceded the light block (Supplementary Figure 5A). Using the same model architecture, we calculated the silhouette score for each rat (Supplementary Figure 5B). However, when the order was reversed (dark preceding light), this discriminability effect disappeared. Thus, while we observed a trend toward higher scores in the dark condition, no statistically significant differences in texture discriminability were observed.

If trial order alone accounted for the increase in discriminability, reversing the order would be expected to yield higher silhouette scores in the light condition. Our findings suggest that

factors related to order (e.g., thirst or motivation, as you proposed) are not the sole contributors. Furthermore, previous studies in human participants have shown that brief blindfolding can produce lingering increases in tactile sensitivity, indicating a lasting effect of visual deprivation. Thus, the absence of significant differences in texture representation when the dark condition preceded the light condition may reflect such lasting effects. We have included a discussion in [lines 441-452].

| *(ii) Discuss explicitly the potential confounding effect of motivational state/thirst.*

We appreciate the reviewer's insightful comment. In the revised manuscript, we now explicitly address the potential confounding role of motivational state and thirst in shaping our results. Because animals were water-restricted to maintain task engagement, it is possible that increasing thirst or fluctuating motivation over the course of a session could alter arousal or attentional state, thereby influencing neural separability. However, when the trial order was reversed (dark condition preceding light), silhouette scores did not show a significant increase in the second (light) trial. Thus, while we acknowledge that motivational state may contribute to trial-to-trial variability, the systematic increase in separability during darkness cannot be fully explained by thirst or motivational confounds. This addition has been incorporated into the discussion section [lines 441-452].

| *(f) Alignment control and the role of forelimb S1.*

| *(i) Repeat the decoding analysis with LFPs aligned to hind-limb strike; report whether the fore-limb dominance persists.*

Thank you for your thoughtful suggestion. We appreciate the opportunity to clarify. Our study was designed to ask a different question: how the absence of visual input reorganizes tactile encoding for the body part that actually initiates texture contact in our paradigm (the forepaw). Accordingly, all analyses were aligned to forelimb strike and our array intentionally oversampled S1-forelimb relative to S1-hindlimb (18 vs. 14 electrodes; Fig. 1F–G), yielding clear topographic forelimb-locked event-related responses (Fig. 3B–D) and forelimb-channel dominance in the decoding explainability analyses (Fig. 5D–E). Repeating the full decoding locked to hind-limb strike would test a different hypothesis and would be difficult to interpret for three reasons:

Design/measurement alignment. Our kinematic detection was built to identify forelimb foot strikes. Extending the detector to hindlimb would require new model training/validation and introduces uncertainty in the exact contact timing relative to the LFP segments we analyze.

Sampling asymmetry. The array and cortical magnification are not balanced across subregions (18 forelimb vs. 14 hindlimb electrodes; Fig. 1G), so a hind-limb-aligned comparison would be confounded by unequal coverage and signal-to-noise across S1 subdivisions rather than reflecting true “dominance.”

Scope of the claim. We do not claim that the forelimb is globally more informative about texture; we show the intuitive and topographically specific result that “forelimb S1 codes textures touching the forelimb,” and that these representations become more separable in darkness (silhouette increase; Fig. 5C). A hind-limb-locked re-analysis would likely reveal hindlimb contributions when the hindpaw is the alignment event — but that would not change the central conclusion about darkness enhancing tactile representational separability.

To address the underlying concern about generality without introducing the above confounds, we have clarified these design choices and limitations in the revised Methods [lines 194-197].

| *(g) Amplitude-based baseline.*

(i) Show that a simple linear discriminant or logistic-regression model on peak amplitudes (and/or other simple features like trough width/slope) cannot reach the CNN's accuracy. This kind of "baseline" analysis could also be useful to pinpoint the discriminative features learned by the CNN.

Thank you for your insightful suggestion. We agree that performing a baseline comparison with a simpler model could help highlight the advantage of using a CNN. However, in our dataset, individual LFP traces do not exhibit clear peaks or well-defined features such as peak amplitude, width, or energy, which makes feature extraction using traditional methods like linear discriminants or logistic regression challenging.

To address this, we performed principal component analysis (PCA) on the raw LFP traces to reduce the dimensionality and applied a support vector machine (SVM) classifier on the reduced features, in line with the approach used for the CNN models (Supplementary Figure 2A). The results of this analysis, demonstrate that the SVM model struggles to effectively discriminate between conditions, further reinforcing the necessity of the CNN model. The CNN's ability to automatically learn complex features from the raw LFP data appears to be a crucial factor in achieving superior classification performance (Supplementary Figure 2B).

(h) Cross-validation and inter-animal generalization.

(i) Consider replacing the single 80/20 split with k-fold cross-validation within animals.

Thank you for this suggestion. Instead of using an 80/20 split, we performed 5-fold cross-validation on all rats. The silhouette scores were averaged within each animal across the five folds, and Figure 6C was updated accordingly. After performing a paired t-test, we still observed a significant difference in silhouette scores between the light and dark conditions.

(ii) Comment on inter-animal generalization.

Thank you for this valuable feedback. Although we did not explicitly test inter-animal generalization, it is unlikely that a model trained on data from one rat would perform equally well when classifying data recorded from another animal. This limitation arises from two main factors. First, despite careful efforts to implant electrodes in the same brain region and cortical layer across experiments, it is impossible to align all 32 electrodes to identical coordinates. Consequently, the recorded LFPs are obtained from slightly different locations, which may reflect distinct neural processing. Second, even within the same species, individual animals differ in brain size and neural circuit organization. Thus, even if electrodes could be placed at identical anatomical locations, inter-individual variability in brain structure would still lead to differences in the recorded signals. Because deep learning models are often sensitive to small perturbations in their input data, we believe that robust inter-animal generalization is unlikely without fine-tuning the model using data from the target animal. This comment has been inserted in the Discussion [lines 494-507].

(2) Writing, figure and terminology improvements (minor):

(a) Figure 5F-G axis label. Decide on either "attribution score" or "activation amplitude" and use that term consistently in panels, legend, and text (currently, I believe it could be confused with raw signal amplitude).

We have unified the terminology to "attribution score" and applied this consistently across the panels, legend, and text.

(b) Throughout the manuscript, use "population-level activity" or "average population dynamics" when discussing LFPs (I believe it is more correct to reserve "population code" for multiple single-unit datasets).

We agree with the reviewer's point and have adapted the term "population dynamics" to describe LFP information consistently throughout the manuscript.

(c) Lines 219-221, state down-sampling to 2 kHz, whereas line 289 mentions 10 kHz. Reconcile these numbers.

We apologize for the confusion and thank the reviewer for thoroughly reading the manuscript. Our original sampling rate was 30 kHz, and all analyses were performed on data resampled to 10 kHz. The reference to 2 kHz was an error, and we have corrected it.

(d) Specify the tail of each statistical test mentioned in the manuscript and any multiple-comparison correction used.

We have specified the tail of each statistical test and any multiple-comparison corrections used in the "Data Analysis" section of the Methods.

(e) Line 244: "variables (He et al., 2015)" → "variables (He et al., 2015)".

We have corrected this formatting issue and revised it to "variables (He et al., 2015)".

(f) Line 253: "one-dimentional" → "one-dimensional".

We have corrected the spelling error and revised it to "one-dimensional".

(3) Data and code sharing:

(a) Consider depositing data and code for the analysis in public open repositories.

Thank you for your suggestion. We have set up a public GitHub repository to share the code. Since the full dataset is quite large (~400GB), we have uploaded a smaller example dataset for the analysis.

<https://doi.org/10.7554/eLife.106554.2.sa0>